MATH 472: INTRODUCTION TO TOPOLOGY

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What is topology?

At its core, topology is the study of continuity, not only continuity of functions, but also continuous deformations of spaces, shapes, and surfaces. The popular conception of topology (Möbius strips, coffee cups = donuts, knots, etc.), however, fails to see what Kyle Ormsby calls the "...stunning breadth and extreme utility of topology in contemporary mathematics." [Orm, Week 1 Lecture Notes]. Here is a woefully incomplete list of places in mathematics and other fields of science where topology arises:

(1) Manifold geometry: the study of manifolds and higher-dimensional surfaces requires passage from local regions to global geometry. Questions about gluing and connection.

(2) Algebraic geometry: the Zariski topology, intersection theory, moduli spaces, complex varieties as topological spaces, etc.

(3) Number Theory: there's a whole set of number-theoretic metrics out there called p-adic metrics, whereby two numbers are close to each other if they share the same divisors. Number theorists do analysis using this measure of distance.

(4) Dynamical Systems and Flows: long-term behavior of differential equations, attractors, and qualitative features that persist under perturbation.

(5) Combinatorics of Geometric Objects: graphs, posets, simplicial complexes; topological invariants constrain which objects exist and how they behave.

(6) Topological Data Analysis: studying high-dimensional point clouds via their topology and using this to create summary statistics about data sets with homology.

(7) Physics: Phase spaces, gauge symmetry, and homotopy theory in high energy physics.

(8) Paleontology (and biology): I just read a book about modern paleontology and lots of the ideas therein seemed decidedly topological (although I doubt the author thought about them like that!). For example, distinguishing species of dinosaur by the number of holes in their skulls, or grouping species based on the number of shared characteristics (this is a notion of distance – hence continuity!)

The breadth of applications here implies that topology is a very abstract subject, and that is true in many ways. That's how topology shows up, but here's some of the ideas that characterize modern topology.

(1) **Properties over construction**: when we study objects that are the same under continuous deformation, it often doesn't matter exactly how they are built. What we want to know is that no matter how you build them, they have the same topological properties. This leads naturally to the idea of category theory and universal properties, where we will *define* objects by the properties they have rather than by their construction.

(2) **Topological equivalence**: there are many different ways to define a circle. Is it the image of $t \mapsto e^{2\pi i t}$ in $\mathbb{C}$? Or is it the image of $\theta \mapsto (\cos\theta, \sin\theta)$ in $\mathbb{R}^2$? Or is it what results from taking the unit interval

$[0, 1]$ and gluing 0 to 1? These are all really the same thing, and we say that they are topologically equivalent. The object that we really want to study is independent of the model chosen. There are multiple ways to measure topological equivalence, some more rigid than others.

(3) **Invariants**: Given many different topologically equivalent avatars for the same object, we need some way to make differentiate between objects even when they're presented differently. Enter invariants: an invariant is a piece of data that stays the same (is "invariant") when replacing an object by a topologically equivalent one. We can use invariants to tell objects apart: if object X has invariant $I(X) = 1$ and object Y has invariant $I(Y) = 0$, then they cannot be topologically equivalent.

Chapter 1

Topological Spaces and Continuous Functions

1.1 Metric Spaces

The first principle above is properties over construction. With that in mind, our first task is to extract the properties of continuity that we learned from analysis and determine which properties determine them. We did this a little bit with the property of distance on the worksheet.

Definition 1.1.1. Let X be a set. A *metric* on X is a function $d\colon X \times X \to [0, \infty)$ with the following properties:

- For any $x \in X$, $d(x, x) = 0$.
- (positivity) For any $x, y \in X$, if $x \neq y$, then $d(x, y) > 0$
- (symmetry) For any $x, y \in X$, $d(x, y) = d(y, x)$
- (triangle inequality) For any $x, y, z \in X$, $d(x, z) \leq d(x, y) + d(y, z)$

Definition 1.1.2. A *metric space* is a pair (X, d) of a set X and a metric d on X.

Example 1.1.3. Here are some examples of metric spaces.

(a) The metric space $(\mathbb{R}^n, d)$ where d is the standard Euclidean metric. For vectors $\vec{u}$ and $\vec{v}$, the distance between them is the length of their difference:

$$d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|.$$

In $\mathbb{R}^2$ with vectors $\vec{u} = (u_0, u_1)$ and $\vec{v} = (v_0, v_1)$, we get

$$d(\vec{u}, \vec{v}) = \sqrt{(u_0 - v_0)^2 + (u_1 - v_1)^2}.$$

(b) The taxicab metric on $\mathbb{R}^n$.

(c) The L^p metric on $\mathbb{R}^n$.

(d) The L^∞ metric on $\mathbb{R}^n$.

(e) The set of binary strings of length n and Hamming distance: the number of positions at which the strings differ.

(f) The discrete metric on any space.

Given a notion of distance, we can talk about open balls in a metric space.

Definition 1.1.4. Let (X, d) be a metric space. Let $x \in X$ and let $r > 0$ be a positive real number. The *open ball of radius* r *around* x is the set

$$B_r(x) := \left\{ y \in X \;\middle|\; d(x, y) < r \right\}.$$

We saw some examples of this on the worksheet.

Example 1.1.5. In the hamming distance example, the open ball of radius 2 around any string is the set of all strings that differ from it by at most one character. The open ball of radius 1 around a given string is just the string itself (remember that it's a strict $<$).

Definition 1.1.6. Let (X, d) be a metric space. An *open set* in X is a set U such that for all $x \in U$, there exists some $r > 0$ such that $B_r(x) \subseteq U$.

Lemma 1.1.7. *The open ball* $B_r(x)$ *is an open set.*

Proof. Let $y \in B_r(x)$. Then $d(x, y) < r$ by the definition of an open ball. Let $\varepsilon = r - d(x, y)$. We claim that $B_\varepsilon(y) \subseteq B_r(x)$. Let $z \in B_\varepsilon(y)$. Then

$$\begin{aligned} d(x, z) &\leq d(x, y) + d(y, z) \\ &< d(x, y) + (r - d(x, y)) \\ &= r. \end{aligned}$$

Therefore, $d(x, z) < r$, so $z \in B_r(x)$. Hence, $B_\varepsilon(y) \subseteq B_r(x)$. □

Example 1.1.8. Draw some pictures of open sets in $\mathbb{R}^2$ with various metrics.

Question 1.1.9. Are there sets in $\mathbb{R}^2$ that are open in the Euclidean metric that are not open in the taxicab metric, or vice versa?

Example 1.1.10. Every subset of $\mathbb{R}^2$ is open for the discrete metric, because the open balls of radius 1 are just points!

1.2 Bases for a Topology

A topology on a set X is the collection of open sets on X (these must follow some rules, which we'll get to soon). For metric spaces, we can define the open sets by those sets U such that around each point $x \in U$, there is an open ball $B_\varepsilon(x)$ completely contained in U. This suggests that the important thing in the topology is not just the open sets, but the open balls in particular. The notion of a basis generalizes the collection of open balls, without any reference to distances.

Recall that if X is a set, then $\mathcal{P}(X)$ is the powerset of X: the set of all subsets of X.

Definition 1.2.1. Let X be a set. A (topological) basis for X is a collection $\beta \subseteq \mathcal{P}(X)$ of subsets of X satisfying the following properties:

- *Covering.* For each $x \in X$, there exists some $B \in \beta$ such that $x \in B$.
- *Filtered.* For all $B_1, B_2 \in \beta$ and all $x \in B_1 \cap B_2$, there exists $B_3 \in \beta$ such that $x \in B_3$ and $B_3 \subseteq B_1 \cap B_2$.

Another way to say the covering property is that the union of all sets in β is X. In symbols:

$$\bigcup_{B \in \beta} B = X.$$

Example 1.2.2. The set of open balls in a metric space forms a basis for that topology. You will prove this on your homework. The proof uses similar ideas to the ones that we used in Lemma 1.1.7.

Example 1.2.3. For $a, b \in \mathbb{Z}$ with $a \neq 0$, let $S(a,b) = \{an + b \mid n \in \mathbb{Z}\}$. This is called an *arithmetic progression*. Then a topological basis on $\mathbb{Z}$ is given by the set of all arithmetic progressions

$$\beta = \left\{ S(a,b) \;\middle|\; a, b \in \mathbb{Z} \right\}.$$

Let's check that this is a basis:

- (covering) Let $n \in \mathbb{Z}$. We must find some arithmetic progression $S(a,b) \in \beta$ such that $n \in S(a,b)$. One option is $S(1,n)$, which contains n because $n = 0(1) + n$.
- (filtered) Suppose that we have basis sets $S(a,b)$ and $S(c,d)$. Let $x \in S(a,b) \cap S(c,d)$. We want to find a basic open set that contains x and is contained in both $S(a,b)$ and $S(c,d)$. The arithmetic progression $S(\text{lcm}(a,c), x)$ is such a basic open set, and $S(\text{lcm}(a,c), x) \subseteq S(a,b) \cap S(c,d)$.

The topology generated by this basis (see below) is called the *Furstenberg topology*. You can use this topology to give a proof of the infinitude of primes different from Euler's proof.

Given a basis, we can define open sets just as we did with the open balls, but with no reference whatsoever to a metric.

Definition 1.2.4. Let X be a set with a basis β. An subset $U \subseteq X$ is *open in* X if for each $x \in U$, there exists some $B \in \beta$ such that $x \in B$ and $B \subseteq U$.

Example 1.2.5. Let X be a set with a topological basis β. Then $\emptyset$ is open (vacuously) and X is open (by the covering axiom).

With the notion of a topological basis, we have axiomatized the idea of open balls to something more general than metric spaces. But we can go even further, and axiomatize the open sets themselves. Let's first think about some of the properties of open sets.

Lemma 1.2.6. *Let X be a set with topological basis β. Let I be a set, and for each $i \in I$, let $U_i \subseteq X$ be an open set in X. Then the union U of all of the U_i,*

$$U := \bigcup_{i \in I} U_i$$

is open as well.

Proof. Let $U = \bigcup_{i \in I} U_i$. We must show that for each $x \in U$, there is some $B \in \beta$ such that $x \in B$ and $B \subseteq U$. To that end, note that if $x \in U$, then $x \in U_j$ for some j. Since U_j is open, there is some $B \in \beta$ such that $x \in B$ and $B \subseteq U_j$. But note that $U_j \subseteq U$ (U_j is a subset of the union of all of the U_is), and therefore this B is the one we search for. In particular, $x \in B$ and $B \subseteq U_j \subseteq U$. □

Lemma 1.2.7. *Let X be a set with topological basis β. Let U and V be open sets in X. Then $U \cap V$ is open.*

Proof. To show that $U \cap V$ is open, we must pick an arbitrary $x \in U \cap V$ and then find some $B \in \beta$ such that $x \in B$ and $B \subseteq U \cap V$.

To that end, let $x \in U \cap V$. Then $x \in U$, and U is open, so there exists some $B_1 \in \beta$ such that $x \in B_1$ and $B_1 \subseteq U$.

Similarly, $x \in V$, and V is open, so there exists some $B_2 \in \beta$ such that $x \in B_2$ and $B_2 \subseteq V$.

Now we have $x \in B_1$ and $x \in B_2$, which means that $x \in B_1 \cap B_2$. By the filtering axiom of a basis, this means that there exists some $B_3 \in \beta$ such that $x \in B_3$ and $B_3 \subseteq B_1 \cap B_2$.

This is the B that we were searching for: since $B_1 \subseteq U$ and $B_2 \subseteq V$, we have

$$B_1 \cap B_2 \subseteq U \cap V.$$

Then $B_3 \subseteq B_1 \cap B_2$ implies that $B_3 \subseteq U \cap V$. We also have $x \in B_3$, so we're done! □

Lemma 1.2.8. *Let X be a set with topological basis β. Let $U_1, \ldots, U_n$ be open sets in X. Then the intersection*

$$\bigcap_{i=1}^{n} U_i$$

is open.

Proof. Proof by induction on n. For $n = 0$ the empty intersection is all of X, which we already showed was open in Example 1.2.5. For $n = 1$, the intersection of one open set is open again. For $n = 2$, this is the previous lemma Lemma 1.2.7. This handles the base cases.

For the inductive step, assume that the intersection of any $n-1$ open sets is open. Let $V = \bigcap_{i=1}^{n-1} U_i$. Then by the previous lemma Lemma 1.2.7, $U_n \cap V$ is the intersection of two opens, and is therefore open. But

$$U_n \cap V = U_n \cap \bigcap_{i=1}^{n-1} U_i = \bigcap_{i=1}^{n} U_i,$$

which is what we were trying to prove is open. □

Example 1.2.9. We can't get away with arbitrary intersections of open sets being open. For example, consider the intersection of all open balls of radius $1/n$ in the Euclidean topology on $\mathbb{R}^2$:

$$\bigcap_{n=1}^{\infty} B_{\frac{1}{n}}((0,0)).$$

This set contains just one point: the origin. And a single point isn't open.

1.3 Topologies

Knowing what we know about the open sets in X with respect to a basis β, we can axiomatize the open sets.

Definition 1.3.1. Let X be a set. A *topology* τ on X is a collection $\tau \subseteq \mathcal{P}(X)$ of subsets of X such that

- Arbitrary unions of sets in τ are again in τ. In symbols: if I is a set and $U_i \in \tau$ for all $i \in I$, then
$$\bigcup_{i \in I} U_i \in \tau.$$
- Finite intersections of sets in τ are again in τ. In symbols: if $U_1, \ldots, U_n \in \tau$ for some nonnegative integer n, then
$$\bigcap_{i=1}^{n} U_i \in \tau.$$

We call the elements of τ the *open sets* of the topology.

Definition 1.3.2. A *topological space* is a pair (X, τ) of a set X and a topology τ on X.

Remark 1.3.3. Some authors also demand the axiom that $\emptyset \in \tau$ and $X \in \tau$. But these follow from the axioms given above! Can you see why?*

This is even more abstract than the abstract bases we talked about earlier! But we already have some examples given by sets with bases.

*The answer is that the union of zero sets is the empty set, so $\emptyset \in \tau$ follows from the unions axiom. Similarly, the intersection of zero sets is all of X, so $X \in \tau$ follows from the finite intersection axiom.

Theorem 1.3.4. *Let X be a set with a topological basis β. Let τ be the set*

$$\tau := \left\{ U \subseteq X \;\middle|\; U \text{ is open with respect to the basis } \beta \right\}.$$

We call this the topology generated by the basis β. *Then τ is a topology on X.*

Proof. This is what we proved in the last section. See Lemma 1.2.6 and Lemma 1.2.8 □

But there are many, many more examples.

Example 1.3.5. On any set X, the *discrete topology* is the topology τ_{disc} where every set is open. Then τ_{disc} is the whole powerset of X: $\tau_{\text{disc}} := \mathcal{P}(X)$

Example 1.3.6. On any set X, the *indiscrete topology* or the *concrete topology* is the topology $\tau_{\text{conc}} := \{\emptyset, X\}$. That is, the only open sets in this topology are empty or the whole set.

Example 1.3.7. The set $\{0, 1\}$ has a topology given by

$$\tau = \left\{ \emptyset,\ \{1\},\ \{0, 1\} \right\}.$$

This is not the discrete topology (because $\{0\}$ isn't open), and it's not the concrete topology (because $\{1\}$ is open). So this is a genuinely new topology on $\{0, 1\}$. Are there any others?

Example 1.3.8. Given a topological space (X, τ) and a subset $A \subseteq X$, we can build a new topology on A by saying that the open sets in A are the sets of the form $U \cap A$ for some open set U of X. In symbols, the topology on A is

$$\tau' = \left\{ U \cap A \;\middle|\; U \in \tau \right\}.$$

This is called the *subspace topology*.

Example 1.3.9. Cofinite topology on $\mathbb{N}$. The topology τ_{cof} is

$$\tau_{\text{cof}} := \left\{ U \subseteq \mathbb{N} \;\middle|\; \mathbb{N} \setminus U \text{ is finite} \right\} \cup \left\{ \emptyset \right\}.$$

In words, the nonempty open sets in this topology are the subsets of $\mathbb{N}$ whose complement is finite. Note that this is different than infinite sets being open! The odd numbers is an infinite set whose complement (the even numbers) is still infinite.

Example 1.3.10. The poset topology. Let $(P, \leq)$ be a poset. A subset $U \subseteq P$ is called *upwards closed* if for all $x \in U$ and $y \in P$ such that $x \leq y$, then $y \in U$ as well. In other words, U is upwards closed if $x \in U$ implies that $y \in U$ for all $y \geq x$.

Let τ be the set of all upwards closed subsets of P. Then τ is a topology on P.

$$\tau = \left\{ U \subseteq P \;\middle|\; U \text{ is upwards closed} \right\}.$$

Frequently in the previous few sections, we made use of open balls around a point. But when we're talking about general topological spaces, we no longer have open balls. So we use the idea of a neighborhood instead. This is very important vocabulary that we will use often!

Definition 1.3.11. Let (X, τ) be a topological space and let $x \in X$. An *(open) neighborhood* of x is an open set $U \in \tau$ such that $x \in U$.

1.4 Closed Sets

Another important idea in topology is the notion of a closed set.

Definition 1.4.1. Let (X, τ) be a topological space. A subset $K \subseteq X$ is *closed* if $X \setminus K$ is open.

Example 1.4.2. Since $\emptyset$ and X are always open, then $X = X \setminus \emptyset$ and $\emptyset = X \setminus X$ are always closed as well. This illustrates that closed and open are not mutually exclusive!

It is false to say that if a set is not open, then it is closed, as the following example shows.

Example 1.4.3. In the Euclidean topology on $\mathbb{R}$, open intervals (a, b) are open sets, and closed intervals $[a, b]$ are closed sets. But there are intervals of the form $[a, b)$ and $(a, b]$ that are neither open nor closed.

Example 1.4.4. In the cofinite topology on $\mathbb{N}$, the closed sets are either finite sets or all of $\mathbb{N}$.

You can also define a topology by specifying the closed sets. In fact, this is probably the better way to define some topologies, like the cofinite topology.

Theorem 1.4.5. *Let X be a set. Let $\kappa \subseteq P(X)$ be a collection of subsets of X such that*

(a) κ is closed under finite unions: if $K_1, \ldots, K_n \in \kappa$, then $\bigcup_{i=1}^{n} K_i \in \kappa$.

(b) κ is closed under arbitrary intersections: if I is a set and $K_i \in \kappa$ for each $i \in I$, then $\bigcap_{i \in I} K_i \in \kappa$.

Then

$$\tau = \left\{ X \setminus K \;\middle|\; K \in \kappa \right\}.$$

is a topology on X.

Proof. To prove that τ is a topology on X, we must show that τ is closed under finite intersections and arbitrary unions.

- (τ is closed under finite intersection) Let $U_1, \ldots, U_n \in \tau$. Because every set in τ is the complement of a set in κ, then for each $i = 1, \ldots, n$ there is some $K_i \in \kappa$ such that $U_i = X \setminus K_i$. Then

$$\bigcap_{i=1}^{n} U_i = \bigcap_{i=1}^{n} (X \setminus K_i) = X \setminus \left(\bigcup_{i=1}^{n} K_i \right).$$

  Since κ is closed under finite union, this shows that the intersection of all the U_is is again a complement of a set in κ. Hence, $\bigcap_{i=1}^{n} U_i \in \tau$.

- (τ is closed under arbitrary union) Let I be a set and let $U_i \in \tau$ for each $i \in I$. Because every set in τ is the complement of a set in κ, then for each $i \in I$, there is some $K_i \in \kappa$ such that $U_i = X \setminus K_i$. Then

$$\bigcup_{i \in I} U_i = \bigcup_{i \in I} (X \setminus K_i) = X \setminus \left(\bigcap_{i \in I} K_i \right).$$

  Since κ is closed under arbitrary intersection, this shows that the union of all of the U_is is again a complement of a set in κ. Hence, $\bigcup_{i \in I} K_i \in \tau$.

This shows that τ is both closed under finite intersection and arbitrary union, and hence τ is a topology. □

1.4.1 Infinitely Many Prime Numbers

We can return to Example Example 1.2.3 and use it to prove that there are infinitely many prime numbers. Recall that we can define a topology on $\mathbb{Z}$ by stating that the basic open sets are the arithmetic progressions $S(a, b)$ with $a \neq 0$, where

$$S(a, b) = \left\{ an + b \;\middle|\; n \in \mathbb{Z} \right\}$$

That is, a set $U \subseteq \mathbb{Z}$ is open in this topology if for all $x \in U$, there is some $S(a, b)$ such that $x \in S(a, b)$ and $S(a, b) \subseteq U$.

Lemma 1.4.6. *$S(a, b)$ is a closed set.*

Proof. We must write $S(a, b)$ as the complement of an open set to show that it is closed.

$$S(a, b) = \mathbb{Z} \setminus \left(\bigcup_{i=1}^{a-1} S(a, b + i) \right)$$

□

This is an example of a topology in which there are many sets that are both open and closed!

Lemma 1.4.7. *Every nonempty open set in this topology is infinite.*

Proof. If U is open and nonempty, then there is some $x \in U$. Since U is open, then there is some $S(a, b)$ such that $x \in S(a, b)$ and $S(a, b) \subseteq U$. Since $S(a, b)$ is an infinite set, then U must be infinite as well. □

Theorem 1.4.8. *There are infinitely many prime numbers.*

Proof. Proof by contradiction. Assume that there are finitely many prime numbers.

By Lemma Lemma 1.4.7, the set $\{-1, +1\}$ cannot be open. Therefore, $\mathbb{Z} \setminus \{-1, +1\}$ cannot be a closed set. However, we can write

$$\mathbb{Z} \setminus \{-1, +1\} = \bigcup_{p \text{ prime}} S(p, 0),$$

which by Lemma Lemma 1.4.6 is a finite union of closed sets, and therefore closed. (Remember we are operating under the assumption of finitely many primes.) This contradicts the fact that $\mathbb{Z} \setminus \{-1, +1\}$ cannot be closed, so our assumption must be false and there are infinitely many primes. □

1.5 Continuous Functions

Topology appears whenever we have a notion of closeness, whether we define it via a metric or a topological basis or a topology.

Definition 1.5.1. Let X and Y be sets, and let $f : X \to Y$ be a function.

(a) Let $B \subseteq Y$. The *preimage* of B under f is the set

$$f^{-1}(B) = \left\{ x \in X \;\middle|\; f(x) \in B \right\}.$$

(b) Let $A \subseteq X$. The *image* of A under f is the set

$$f(A) = \left\{ f(a) \;\middle|\; a \in A \right\}.$$

Definition 1.5.2. Let X and Y be topological spaces, and let $f : X \to Y$ be a function. The function f is *continuous* if for each open subset V of Y, $f^{-1}(V)$ is open.

Let's see how this definition of continuity matches up with the definition of continuity you learned in analysis.

Proposition 1.5.3. *Let $f: \mathbb{R} \to \mathbb{R}$ be a continuous function, meaning that for all $x_0 \in \mathbb{R}$ and all $\varepsilon > 0$, there is some $\delta > 0$ such that if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \varepsilon$.*

Then f is continuous in the sense of topology (Definition 1.5.2).

Proof. Assume that $f: \mathbb{R} \to \mathbb{R}$ is a continuous function. Let $V \subseteq \mathbb{R}$ be an open set. We want to show that $f^{-1}(V)$ is open.

To show that $f^{-1}(V)$ is open, we must pick some point $x_0 \in f^{-1}(V)$ and show that we can fit an open ball around x_0 entirely inside $f^{-1}(V)$.

Since V is open, we can fit an open ball around any point inside of V. In particular, $f(x_0) \in V$ by the definition of $x_0 \in f^{-1}(V)$. Therefore, there is an open ball $B_\varepsilon(f(x_0)) \subseteq V$.

The analysis definition of continuity shows us that if $x \in B_\delta(x_0)$, then $f(x) \in B_\varepsilon(f(x_0))$. In other words,

$$B_\delta(x_0) \subseteq f^{-1}(B_\varepsilon(f(x))).$$

Since $B_\varepsilon(f(x_0)) \subseteq V$, we see then that $f^{-1}(B_\varepsilon(f(x_0))) \subseteq f^{-1}(V)$. Therefore,

$$B_\delta(x_0) \subseteq f^{-1}(B_\varepsilon(f(x))) \subseteq f^{-1}(V).$$

This finds an open set around x_0 that is contained in $f^{-1}(V)$. Since our choice of x_0 was arbitrary, $f^{-1}(V)$ is open. Therefore, f is continuous in the sense of topology. □

Exercise 1.5.4. Prove that if $f: \mathbb{R} \to \mathbb{R}$ is continuous in the sense of topology (Definition 1.5.2), then $f: \mathbb{R} \to \mathbb{R}$ is continuous in the sense of analysis.

Example 1.5.5.

- identity functions are continuous
- every function whose domain is discrete is continuous
- from X one topology to X with another topology
- taxicab geometry and regular geometry

1.5.1 Building Continuous Functions

How can you go about building continuous functions from one space to another? There are a couple of standard techniques.

Proposition 1.5.6. *Let X and Y be topological spaces and let $y \in Y$. Then the constant function $f: X \to Y$ defined by $f(x) = y$ for all $x \in X$ is continuous.*

Proof. Let $V \subseteq Y$ be open. If $y \in V$, then $f^{-1}(V) = X$, which is open. If $y \notin V$, then $f^{-1}(V) = \emptyset$, which is also open. Therefore, the preimage of any open in Y is open, which means that f is continuous. □

Proposition 1.5.7. *Let X, Y, and Z be topological spaces with continuous functions $f: X \to Y$ and $g: Y \to Z$. Then $g \circ f: X \to Z$ is continuous.*

Proof. To show that $g \circ f: X \to Z$ is continuous, we need to show that for each open set $W \subseteq Z$, $(g \circ f)^{-1}(W)$ is open in X. But notice that

$$(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W)).$$

Since g is continuous and W is open, then $g^{-1}(W)$ is open. Since f is continuous and $g^{-1}(W)$ is open, then $f^{-1}(g^{-1}(W))$ is open. Therefore, $(g \circ f)^{-1}(W)$ is open, so $g \circ f$ is continuous. □

Lemma 1.5.8. *Let X be a topological space with a subspace A. Let $i\colon A \hookrightarrow X$ be the inclusion function. Then $i\colon A \to X$ is continuous.*

Proof. Let $U \subseteq X$ be open. Then $i^{-1}(U) = U \cap A$, which is open in A by the definition of the subspace topology. □

Proposition 1.5.9. *Let $f\colon X \to Y$ be a continuous function between topological spaces. Let $A \subseteq X$ be a subspace with the subspace topology. Then if we restrict the domain of f to A, it is still continuous.*

Proof. Let $i\colon A \hookrightarrow X$ be the inclusion function. Then restricting the domain of f to A is the same as composing $f \circ i$. Since f is continuous (by assumption) and i is continuous (by previous lemma), then $f \circ i$ is continuous by a previous proposition. □

Lemma 1.5.10. *Let X be a topological space with subspace A. Give A the subspace topology.*

(a) If A is open in X and $U \subseteq A$ is open in A, then U is open in X.

(b) If A is closed in X and $K \subseteq K$ is closed in A, then K is closed in X.

Theorem 1.5.11 (Pasting)**.** *Let X be a topological space such that $X = A \cup B$, where A and B are closed in X. Then let $f\colon A \to Y$ and $g\colon B \to Y$ be continuous functions, where A and B are given the subspace topology. If $f(x) = g(x)$ for all $x \in A \cap B$, then f and g can be combined to give a continuous function $h\colon X \to Y$ defined by*

$$h(x) = \begin{cases} f(x) & \text{if } x \in A \\ g(x) & \text{if } x \in B. \end{cases}$$

Proof. We will use the formulation of continuity where $f\colon Y \to Z$ is continuous if and only if for each closed subset $K \subseteq Z$, $f^{-1}(K)$ is closed in Y.

Let C be a closed set in Y. Then $h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$. Since f is continuous, $f^{-1}(C)$ is closed in A, and therefore closed in X by the previous lemma. Since g is continuous, $g^{-1}(C)$ is closed in B, and therefore closed in X by the previous lemma. Hence, $h^{-1}(C)$ is the union of two closed sets, and therefore closed. So h sends closed sets to closed sets and is therefore continuous. □

Example 1.5.12. The function

$$h(x) = \begin{cases} 2x & \text{if } \ x \leq 0 \\ x/2 & \text{if } \ x \geq 0 \end{cases}$$

is defined by two functions $f\colon (-\infty, 0] \to \mathbb{R}$, $f(x) = 2x$ and $g\colon [0, \infty) \to \mathbb{R}$, $g(x) = x/2$ on two closed sets $A = (-\infty, 0]$ and $B = [0, \infty)$. The functions f and g agree on $A \cap B = \{0\}$. The pasting theorem guarantees that h is continuous.

1.5.2 Homeomorphisms

Definition 1.5.13. A function $f\colon X \to Y$ between topological spaces is a *homeomorphism* if

- f is continuous
- f is a bijection with inverse function g
- the inverse function $g\colon Y \to X$ is continuous

Definition 1.5.14. If $f\colon X \to Y$ is a homeomorphism, then we say that X and Y are *homeomorphic* and write $X \cong Y$.

Example 1.5.15. The open interval $(0,1)$ is homeomorphic to any other open interval (a,b) via

$$f(x) = xb + (1-x)a = a + x(b-a).$$

The inverse is

$$g(x) = \frac{x-a}{b-a}.$$

Example 1.5.16. The closed interval $[0,1]$ is homeomorphic to $[a,b]$ via the same formulas above.

Example 1.5.17. The open interval $(0,1)$ is homeomorphic to $\mathbb{R}$ via

$$f(x) = \frac{1}{x(x-1)}.$$

Example 1.5.18. Stereographic projection. The unit sphere S^2 is the set of points a distance of 1 from the origin in $\mathbb{R}^3$.

$$S^2 = \left\{ (x,y,z) \in \mathbb{R}^3 \;\middle|\; x^2+y^2+z^2 = 1 \right\}.$$

Let $N = (0,0,1)$ be the north pole of S^2. Then $S^2 \setminus \{N\}$ is homeomorphic to $\mathbb{R}^2$ via the functions $f\colon S^2 \setminus \{N\} \to \mathbb{R}^2$

$$f(x,y,z) = \left(\frac{x}{1-z}, \frac{y}{1-z} \right),$$

and $g\colon \mathbb{R}^2 \to S^2 \setminus \{N\}$

$$g(x,y) = \left(\frac{2x}{1+x^2+y^2}, \frac{2y}{1+x^2+y^2}, \frac{-1+x^2+y^2}{1+x^2+y^2} \right).$$

You can check that these are inverse functions and are continuous. There's also a nice geometric description of this process: for a point $P = (x,y,z)$ on $S^2 \setminus \{N\}$, draw the straight line in $\mathbb{R}^3$ through N and P. This will intersect the plane $(z=0)$ in one point, which is the point $f(P)$ in $\mathbb{R}^2$. Conversely, given a point in the plane $(z=0)$, you can draw a line through it and N to find a unique point on the sphere.

It's very important that the inverse function is also continuous! It's not automatic that the inverse is continuous, as the following example shows.

Example 1.5.19. $[0,1) \to S^1$ given by $t \mapsto e^{2\pi i t}$ is continuous and a bijection, but no continuous inverse. In particular, $[0, \frac{1}{2})$ is open in the interval $[0,1)$ under the subspace topology but its preimage in the circle S^1 is not open in the subspace topology on $\mathbb{C}$; it includes $z=1$ along the real axis.

Definition 1.5.20. Let $f\colon X \to Y$ be a function between topological spaces. Then we say that f is

- *open* if for each open $U \subseteq X$, $f(U)$ is open in Y.
- *closed* if for each closed $K \subseteq X$, $f(K)$ is closed in Y.

There's another useful characterization of homeomorphisms.

Lemma 1.5.21. *Let $f\colon X \to Y$ be a bijection with inverse $g\colon Y \to X$, and let $A \subseteq X$ be any subset. Then*

$$g^{-1}(A) = f(A).$$

Proof.

$$g^{-1}(A) = \{y \in Y \mid g(y) \in A\} = \{f(g(y)) \in Y \mid g(y) \in A\} = \{f(x) \in Y \mid x \in A\} = f(A).$$

□

Proposition 1.5.22. *Let $f\colon X \to Y$ be a continuous function between topological spaces. Then the following are equivalent:*

(a) $f\colon X \to Y$ *is a homeomorphism*

(b) $f\colon X \to Y$ *is an open bijection*

Proof. ($\Longleftarrow$). Assume that $f\colon X \to Y$ is an open bijection. Let $g\colon Y \to X$ be the inverse function of f. To show that f is a homeomorphism, we must show that $g\colon Y \to X$ is continuous.

To do so, let $U \subseteq X$ be open. We want to show that $g^{-1}(U)$ is open in Y. But notice that by the lemma, $g^{-1}(U) = f(U)$, which is open because f is an open function.

($\Longrightarrow$). Assume that $f\colon X \to Y$ is a homeomorphism, and let $g\colon Y \to X$ be the inverse function to f. To show that f is an open function, we must show that for any $U \subseteq X$, $f(U)$ is open. But by the lemma, $f(U) = g^{-1}(U)$, which is open because g is continuous. □

This shows that a homeomorphism $f\colon X \to Y$ not only sends points in X bijectively to points in Y, but also open sets in X bijectively to open sets in Y. For that reason, whenever there is any property of X that can be expressed entirely in terms of the topology of X (that is, entirely in terms of the open sets of X), then that property applies to Y as well. Such a property of X is called a *topological property*.

Definition 1.5.23. A *topological property* P is a property of topological spaces such that a space X has property P and $X \cong Y$ implies that Y has property P as well.

In other words, a topological property is a property that is *homeomorphism invariant*.

1.6 Comparing Topologies

This is a good time to bring up some vocabulary that comes up when comparing topologies on the same set.

Definition 1.6.1. Suppose that X is a set and that τ, τ' are two topologies on X.

(a) We say that τ' is *finer* than τ if $\tau' \supseteq \tau$.

(b) We say that τ' is *coarser* than τ if $\tau' \subseteq \tau$.

This vocabulary is meant to convey the analogy of gravel or dirt. If you think of the open sets of a topology as pebbles, and you smash them into bits, then the collection of open sets has been enlarged and the topology is said to have been made finer. So even though the term *finer* suggests some idea of "smaller-than," the *larger* topology τ' is the finer one (because it has smaller open sets).

To be clear:

- If τ' is finer than τ, then every set that is open in τ is open in τ'.
- If τ' is coarser than τ, then every set that is open in τ' is open in τ.

Example 1.6.2. The discrete topology on a set X is finer than any other topology on X; it is the finest topology.

Example 1.6.3. The indiscrete topology on a set X is coarser than any other topology on X; it is the coarsest topology.

Example 1.6.4. The *lower limit topology* on $\mathbb{R}$ is the topology on $\mathbb{R}$ with a basis of open sets given by $[a, b)$. Let $\mathbb{R}_\ell$ denote the topological space that is the set $\mathbb{R}$ equipped with the lower limit topology, and let $\mathbb{R}_s$ denote $\mathbb{R}$ with the standard topology.

In $\mathbb{R}_\ell$, (a, b) is open because

$$(a, b) = \bigcup_{n=1}^{\infty} \left[a + \tfrac{1}{n}, b\right).$$

But $[a, b)$ is not open in $\mathbb{R}_s$. Therefore, $\mathbb{R}_\ell$ has more open sets, so it is finer than $\mathbb{R}_s$.

1.7 The Product Topology

How is the topology on $\mathbb{R}$ related to the topology on $\mathbb{R}^2$? One answer is that we can think of $\mathbb{R}$ as a subset of $\mathbb{R}^2$ (as the x-axis, say, or as any line), and that the subspace topology on this line gives the usual topology on $\mathbb{R}$. Another answer is that we can build the standard topology on $\mathbb{R}^2$ using the topology on $\mathbb{R}$ as a product topology.

Definition 1.7.1. Let X and Y be topological spaces. The *product topology* on $X \times Y$ is the topology with basis $U \times V$, where U is open in X and V is open in Y.

We also apply this definition to products of finitely many spaces.

Definition 1.7.2. Let $X_1, \ldots, X_n$ be topological spaces. The *product topology* on $X_1 \times X_2 \times \cdots \times X_n$ is the topology with basis given by sets of the form $U_1 \times U_2 \times \cdots \times U_n$, where U_i is open in X_i for each $i = 1, 2, \ldots, n$.

Example 1.7.3. Consider the product topology on $\mathbb{R} \times \mathbb{R}$, where $\mathbb{R}$ has the standard topology. A basic open set in the product topology is an open box of the form $(a, b) \times (c, d)$, and a set U is open in this topology if and only if for all $x \in U$, there is some open box B such that $x \in B$ and $B \subseteq U$.

This space is homeomorphic to $\mathbb{R}^2$ with the standard topology (wherein the basic open sets are open balls).

There are two important functions for any product $X \times Y$.

Definition 1.7.4. Let X and Y be sets (or spaces). The *projection functions* are the functions $p_1 : X \times Y \to X$ and $p_2 : X \times Y \to Y$ defined by

$$p_1(x, y) = x$$
$$p_2(x, y) = y.$$

When X and Y are topological spaces, these functions have nice topological properties.

Proposition 1.7.5. *Let X and Y be spaces and give $X \times Y$ the product topology. Then the projections p_1 and p_2 are continuous and open.*

Proof. We will prove this for p_1; the case for p_2 is basically the same.

To show that $p_1 : X \times Y \to X$ is continuous, we want to show that for any $U \subseteq X$ open, $p_1^{-1}(U)$ is open in $X \times Y$. But $p_1^{-1}(U) = U \times Y$, which is open in $X \times Y$ since U and Y are both open in X and Y, respectively. Hence, p_1 is continuous.

To show that $p_1 : X \times Y \to X$ is open, we must show that for any open set $W \subseteq X \times Y$, $p_1(W)$ is open in X. It suffices to show this when W is a basic open set (exercise). If W is a basic open set, then $W = U \times V$ where U is open in X and V is open in Y. Then

$$p_1(W) = p_1(U \times V) = U,$$

which is open in X. □

The projection functions help us find continuous functions into the product.

Proposition 1.7.6. *Let X, Y, and Z be topological spaces. A function $f : Z \to X \times Y$ is continuous if and only if both $p_1 \circ f$ and $p_2 \circ f$ are continuous.*

Proof. ($\Longrightarrow$). If f is continuous, then $p_1 \circ f$ and $p_2 \circ f$ are both continuous by Proposition 1.5.7.

($\Longleftarrow$). To show that f is continuous, we must show that $f^{-1}(U \times V)$ is open in Z for each basic open set $U \times V$ of $X \times Y$ by a homework problem.

$$f^{-1}(U \times V) = (p_1 \circ f)^{-1}(U) \cap (p_2 \circ f)^{-1}(V),$$

and since $p_1 \circ f$ and $p_2 \circ f$ are both continuous, then the right hand side is open. Therefore, $f^{-1}(U \times V)$ is open in Z. □

There is a neat characterization of the product topology in terms of the projections.

Proposition 1.7.7. *The product topology on $X \times Y$ is the coarsest topology on $X \times Y$ for which both p_1 and p_2 are continuous.*

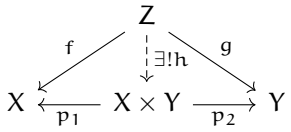
Proof. Suppose that τ is any topology on $X \times Y$ topology such that both projections are continuous with respect to τ. Then for open sets $U \subseteq X$ and $V \subseteq Y$, $p_1^{-1}(U) \cap p_2^{-1}(Y)$ must be open in $X \times Y$. But

$$p_1^{-1}(U) \cap p_2^{-1}(Y) = U \times V.$$

Therefore, in *any* topology such that both projections are continuous, $U \times V$ must be open for any open $U \subseteq X$ and $V \subseteq Y$. So any such topology contains the basic open sets of the product topology, and so τ must be finer than the product topology. In other words, the product topology is coarser than τ. □

We can combine the previous two propositions into the following.

Theorem 1.7.8 (Universal Property of the Product)**.** *Let X and Y be spaces. The product $X \times Y$ is the topological space with two functions $p_1 : X \times Y \to X$ and $p_2 : X \times Y \to Y$ such that for any other topological space Z and continuous functions $f : Z \to X$ and $g : Z \to Y$, there is a unique continuous function $h : Z \to X \times Y$ such that $p_1 \circ h = f$ and $p_2 \circ h = g$.*



Chapter 2

Topological Properties

2.1 Categories

Category theory was introduced by Samuel Eilenberg and Saunders Mac Lane in the mid 1900s to put their work on algebraic topology on solid ground. Although it was originally motivated by topological considerations, category theory has become an incredibly useful tool in many areas of mathematics.

Category theory builds on and formalizes a philosophical shift that occurred in mathematics in the 1900s, namely that we should think not only about mathematical objects themselves, but also about the relations between them. These relations take many forms, but often they take the form of functions between sets with some kind of special property. Once we have the language of category theory, we can prove theorems about the categories themselves and apply them to different areas of mathematics systematically. In fact, many seemingly different ideas in different branches of math turn out to come from the same categorial constructions. In other words, category theory becomes a meta-mathematical tool that we can use to translate results from one area of mathematics to another.

We begin with a motivating example.

Example 2.1.1. One kind of mathematical object that shows up everywhere is sets. Sets are related by functions. Any function $f\colon A \to B$ has a domain A (also called the source) and a codomain B (also called the target). You can also compose functions: given $f\colon A \to B$ and $g\colon B \to C$, there is a new function called $g \circ f\colon A \to C$ given by $a \mapsto g(f(a))$ for all $a \in A$. Composition of functions is associative: if we have a third function $h\colon C \to D$, we have

$$(h \circ g) \circ f = h \circ (g \circ f).$$

For every set A, there is an identity function $\mathrm{id}_A : A \to A$ given by $\mathrm{id}_A(a) = a$. The special thing about the identity function is that for any function $f\colon A \to B$,

$$f \circ \mathrm{id}_A = f,$$

and for any function $k\colon E \to A$,

$$\mathrm{id}_A \circ k = k.$$

It is this kind of thing that the definition of a category seeks to generalize.

Definition 2.1.2. A *category* $\mathcal{C}$ is the following data:

- A collection* of *objects* $\mathrm{Ob}(\mathcal{C})$.

*We say a *collection* instead of *set* because we will consider things like the category of all sets. But there is no set of all sets because that creates paradoxes (does the set of all sets contain itself?). So we say that we're allowed to have a collection of all sets, and that a collection is not a set, to sidestep this issue. For the most part, you never need to worry about this and you can just leave it to the logicians to sort it out.

- A collection of *morphisms* $\mathrm{Mor}(\mathcal{C})$ such that each morphism $f \in \mathrm{Mor}(\mathcal{C})$ has both a domain object and a codomain object. If $X, Y \in \mathrm{Ob}(\mathcal{C})$, we write $\mathcal{C}(X,Y)$[†] for the collection of morphisms with domain X and codomain Y. If $f \in \mathcal{C}(X,Y)$, we will also write $f\colon X \to Y$.
- For each $X \in \mathrm{Ob}(\mathcal{C})$, there is an *identity morphism* $\mathrm{id}_X\colon X \to X$.
- For any three objects $X, Y, Z \in \mathrm{Ob}(\mathcal{C})$, there is a composition operation

$$\circ\colon \mathcal{C}(Y,Z) \times \mathcal{C}(X,Y) \to \mathcal{C}(X,Z),$$

  that composes morphisms $g\colon Y \to Z$ and $f\colon X \to Y$ to form a new morphism $g \circ f\colon X \to Z$.

These data must satisfy the following properties:

- *Associativity*. Composition is associative. In more detail, given $f\colon X \to Y$, $g\colon Y \to Z$, and $h\colon Z \to W$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

- *Identity*. For any $f\colon X \to Y$, we have

$$f \circ \mathrm{id}_X = f \qquad \text{and} \qquad \mathrm{id}_Y \circ f = f.$$

While it looks like a lot to write down, the thing to remember about categories is this: there's objects and morphisms and composition of morphisms, and the composition works exactly like you'd expect if it were composition of functions. You've been working with categories for your whole career as mathematicians, without realizing it.

Example 2.1.3. We've already mentioned that there is a category $\mathcal{S}\mathrm{et}$ of sets and functions. Similarly, there is also a category $\mathcal{F}\mathrm{in}$, where the objects are finite sets and the morphisms are functions.

Example 2.1.4. The category $\mathcal{V}\mathrm{ect}_{\mathbb{R}}$ has objects $\mathbb{R}$-vector spaces and morphisms linear transformations. Since linear transformations compose to give new linear transformations, and the identity functions $\mathrm{id}_V\colon V \to V$ are linear transformations, this is well-defined as a category. There is also a category $\mathcal{V}\mathrm{ect}_{\mathbb{C}}$ of $\mathbb{C}$-vector spaces.

We've been working in one particular category this entire class.

Example 2.1.5. The objects of the category $\mathcal{T}\mathrm{op}$ are topological spaces. The morphisms in this category are continuous functions. This category is well-defined because composing two continuous functions gives another continuous function, and identity functions are continuous.

There are also examples that are less familiar.

Example 2.1.6. The empty category $\emptyset$ has no objects (i.e. $\mathrm{Ob}(\emptyset) = \emptyset$) and no morphisms ($\mathrm{Mor}(\emptyset) = \emptyset$). Composition is the empty function $\emptyset \to \emptyset$ and all properties are satisfied vacuously.

Example 2.1.7. The trivial category $\mathbf{1}$ has only one object $\mathrm{Ob}(\mathbf{1}) = \{*\}$, and a single morphism $\mathrm{id}_*\colon * \to *$. The only composition to define is $\mathrm{id} \circ \mathrm{id} = \mathrm{id}$, and this gives the associative property and identity property too.

[†]You may have seen $\mathrm{Hom}_{\mathcal{C}}(X,Y)$ in other classes for the same thing.

2.1.1 Commutative Diagrams

It is useful to draw morphisms in a category $\mathcal{C}$ diagrammatically. We may write $f\colon X \to Y$ to mean $f \in \mathcal{C}(X, Y)$, but we may also draw

$$X \xrightarrow{f} Y$$

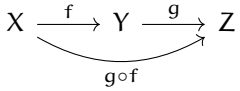
to mean the same information. Now suppose we have another morphism

$$Y \xrightarrow{g} Z.$$

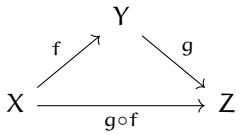
Then we can compose these to get a morphism

$$X \xrightarrow{g \circ f} Z.$$

We can put all of this information into a single *commutative diagram*

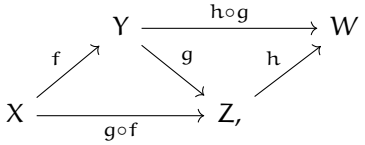


The exact geometry of the diagram doesn't matter. It could equally well be drawn as a triangle

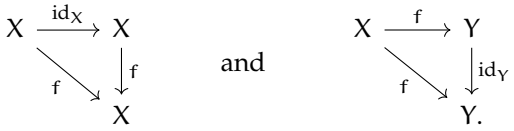


and it would convey the same information.

We can use commutative diagrams to communicate the axioms of a category. For example, associativity is the diagram

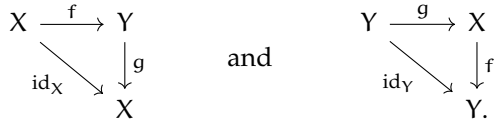


which says that we can paste together commutative diagrams to get commutative quadrilaterals. The identity axiom becomes



2.1.2 Isomorphisms

Definition 2.1.8. Let $\mathcal{C}$ be a category. A morphism $f\colon X \to Y$ in $\mathrm{Mor}(\mathcal{C})$ is an *isomorphism* if there exists a morphism $g\colon Y \to X$ in $\mathrm{Mor}(\mathcal{C})$ such that the following diagrams commute:



In equations, these diagrams are $f \circ g = \mathrm{id}_X$ and $g \circ f = \mathrm{id}_Y$. In other words, $f\colon X \to Y$ is an isomorphism if it has a two-sided inverse $g\colon Y \to X$. If there is an isomorphism $f\colon X \to Y$, then we say that X and Y are *isomorphic* and write $X \cong Y$.

Example 2.1.9. In the category $\mathcal{S}et$, a morphism $f\colon X \to Y$ is an isomorphism if and only if it is a bijection.

Example 2.1.10. In the category $\mathcal{V}ect_{\mathbb{R}}$, a linear transformation $T\colon V \to W$ is an isomorphism if and only if it is a bijective linear transformation.

It might be tempting to say that the isomorphism in $\mathcal{T}op$ are continuous bijections, but we've already seen counterexamples to that.

Example 2.1.11. In the category $\mathcal{T}op$, a morphism $f\colon X \to Y$ is an isomorphism if and only if it is a homeomorphism.

This explains why we insist that homeomorphisms have a continuous inverse. By the definition of an isomorphism, the inverse function must live in the same category as the function itself!

2.2 Universal Properties

Universal properties are a way to understand constructions in a category in terms of the morphisms into or out of that construction. Because the particular construction depends on the category that you work in, the universal properties prefer an "external" description in terms of morphisms. In other words, the universal property specifies an object by some properties that it should have.

When we study objects by universal properties, we will follow a consistent blueprint:

(1) Universal property. Say what we want the object to do in terms that apply to many different categories.

(2) Construction. Provide a concrete description of the object in our category of choice, and verify that it satisfies the universal property.

(3) Examples. Get familiar with the construction and universal property by studying instances of it in a category of our choice.

(4) Other contexts. Once we are familiar with our object and its universal property in one category (usually $\mathcal{T}op$, we can apply it in another category to build bridges between different areas of math.

2.2.1 Products

Given objects X and Y in a category $\mathcal{C}$, we can consider the collection of all diagrams

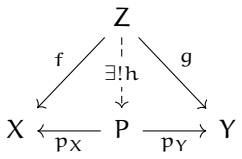
$$X \longleftarrow Z \longrightarrow Y$$

in $\mathcal{C}$. In other words, we are considering objects $Z \in \mathrm{Ob}(\mathcal{C})$ which admit two morphisms with codomains X and Y. A typical category theory question is whether or not there is a "universal" object mapping to both X and Y. Such a universal object is referred to as the product of X and Y in $\mathcal{C}$.

Definition 2.2.1. Let $P \in \mathrm{Ob}(\mathcal{C})$, and suppose there are morphisms $p_X \in \mathcal{C}(P, X)$ and $p_Y \in \mathcal{C}(P, Y)$ such that for any $f \in \mathcal{C}(Z, X)$ and $g \in \mathcal{C}(Z, Y)$, there is a unique morphism $h \in \mathcal{C}(Z, P)$ such that $p_X \circ h = f$ and $p_Y \circ h = g$. The data of P together with the morphisms p_X and p_Y is called the *product* of X and Y in $\mathcal{C}$. We often denote the object P by $X \times Y$.

Frequently, we will abuse notation and say that $P = X \times Y$ is the product of X and Y, and leave the morphisms p_X and p_Y implicit.

We can summarize the definition in one commutative diagram (which doesn't tell you everything about the product, but is convenient visually):



The arrow h labelled $\exists!$ means that there exists ($\exists$) a unique (!) morphism which makes the diagram commute.

Question 2.2.2.

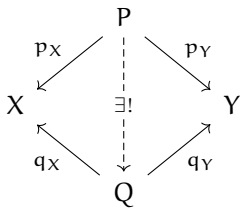
(a) Does the product of X and Y exist?

(b) If it does exist, is it unique?

To answer (a), notice that this definition isn't constructive. It tells you about a property that an object should have. This raises the question of whether or not the product always exists. The answer to this question is no, and there are examples to that effect.

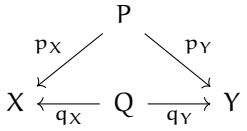
Example 2.2.3. Let Σ be the category with objects $[n] := \{1, 2, \ldots, n\}$, and $\Sigma([n], [m])$ is the set of bijections from $[n]$ to $[m]$. Note that when $n \neq m$, $\Sigma([n], [m]) = \emptyset$. In order for the product to exist in a category $\mathcal{C}$, we must have morphisms $X \leftarrow X \times Y$ and $X \times Y \rightarrow Y$. In the category Σ, this will not happen unless $X = X \times Y$!

The answer to (b), however, is yes. Namely, when it exists, $X \times Y$ is *unique up to unique isomorphism*, meaning that if there are two products $X \xleftarrow{p_X} P \xrightarrow{p_Y} Y$ and $X \xleftarrow{q_X} Q \xrightarrow{q_Y} Y$ in $\mathcal{C}$ both satisyfing the definition of the product, then there is a unique isomorphism $P \rightarrow Q$ such that the diagram below commutes:

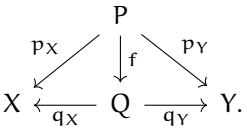


Proposition 2.2.4. *If the product $X \times Y$ exists in $\mathcal{C}$, then it is unique up to unique isomorphism in the sense above.*

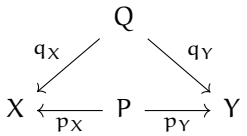
Proof. Suppose that $X \xleftarrow{p_X} P \xrightarrow{p_Y} Y$ and $X \xleftarrow{q_X} Q \xrightarrow{q_Y} Y$ both satisfy the definition of a product in $\mathcal{C}$. Then the diagram



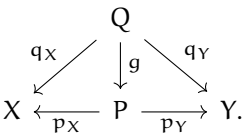
and the universal property of Q tell us that there is a unique morphism $f\colon P \rightarrow Q$ such that the following diagram commutes:



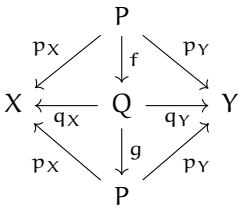
We must show that f is an isomorphism. To that end, consider the diagram



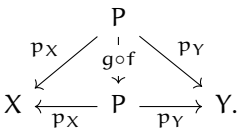
and use the definition of the product for P to produce a unique morphism $g\colon Q \rightarrow P$ such that the following diagram commutes:



Putting these commutative diagrams together, we get



In particular, erasing the Q from this diagram shows us that the following commutes:



But note that putting $1_P \colon P \to P$ in place of $g \circ f$ in the diagram above also makes this diagram commute. Since the definition of the product for P insists that the morphism that fits into the diagram is unique, this means that $1_P = g \circ f$.

We can also work out a similar argument to show that $f \circ g = 1_Q$. Therefore, $P \cong Q$, and we have proven that the product is unique up to unique isomorphism. □

We now see if our definition of product matches up with our ideas about products in various categories.

Proposition 2.2.5. *In the category* $\mathcal{S}\mathrm{et}$*, the cartesian product*

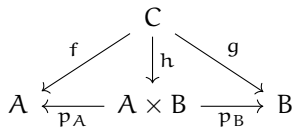
$$A \times B = \left\{ (a, b) \;\middle|\; a \in A, b \in B \right\}$$

along with the projections $p_A \colon A \times B \to A$, $p_A(a, b) = a$ *and* $p_B \colon A \times B \to B$, $p_B(a, b) = b$ *is the categorical product of* A *and* B*.*

Proof. Given any set C with functions $A \xleftarrow{f} C \xrightarrow{g} B$, we define $h \colon C \to A \times B$ by

$$h(c) = \Big(f(c), g(c) \Big).$$

It is an exercise to check that the diagram



commutes and that h is the only function that would make this diagram commute. Therefore, $A \times B$ is indeed the categorical product of A and B. □

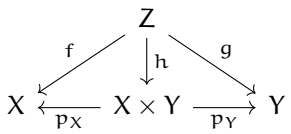
Knowing only the universal property of the product, how might you figure out what the product topology on $X \times Y$ should be? To start, we know that the product $X \times Y$ comes equipped with projection morphisms $p_X \colon X \times Y \to X$ and $p_Y \colon X \times Y \to Y$, and these must be morphisms in the category $\mathcal{T}\mathrm{op}$. Therefore, p_X and p_Y must be continuous functions, so $p_X^{-1}(U)$ and $p_Y^{-1}(V)$ must be open for any open sets $U \subseteq X$ and $V \subseteq Y$. Moreover, open sets in any topology must be closed under finite intersection, so

$$U \times V = p_X^{-1}(U) \times p_Y^{-1}(V)$$

must be open in $X \times Y$ as well. Since the universal property of a product doesn't seem to put any additional restrictions on the open sets, our best guess is that the topology on $X \times Y$ is the one generated by the basic open sets $U \times V$. This turns out to be the correct guess.

Proposition 2.2.6. *Let X and Y be topological spaces, and let τ be the topology on $X \times Y$ generated by the basis of open sets $U \times V$ with U open in X and V open in Y. Then $(X \times Y, \tau)$ equipped with the standard projection maps is the categorical product of X and Y.*

Proof. We check that the set $X \times Y$, equipped this topology, satisfies the universal property of the product. Let Z be a topological space and let $f\colon Z \to X$ and $g\colon Z \to Y$ be continuous functions. From Proposition 2.2.5, we know that the function $h\colon Z \to X \times Y$ defined by $h(z) = \big(f(z), g(z)\big)$ is the unique function that makes the following diagram commute (as a diagram in the category $\mathcal{S}et$):



It remains to show that h is a continuous function. Since the topology τ on $X \times Y$ is generated by a basis of sets $U \times V$, it suffices to check that $h^{-1}(U \times V) \subseteq Z$ is open. But $h(z) = \big(f(z), g(z)\big)$, so

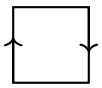
$$h^{-1}(U \times V) = f^{-1}(U) \cap g^{-1}(V).$$

Since both f and g are continuous and finite intersections of open sets are open, we see that $h^{-1}(U \times V)$ is open, and therefore continuous. □

2.2.2 Quotient Topology

Quotients are ubiquitous in mathematics. You may be familiar with the example of modular arithmetic, which studies not $\mathbb{Z}$, but the quotient $\mathbb{Z}/n\mathbb{Z}$. To form this quotient, we identify two integers $x, y \in \mathbb{Z}$ if and only if $x - y = n$.

In topology, we are interested in taking topological spaces and gluing pieces together to form new spaces. For example, consider a closed unit square in $\mathbb{R}^2$ with the subspace topology. If we take the two vertical edges of the square and identify them with each other only after reversing one, and then we glue them together, we end up with a Möbius strip.

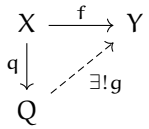


This is the kind of construction that the quotient topology is supposed to describe. We will study it first via the universal property, and then give a construction, and then do many examples.

Definition 2.2.7. Let X be a set with an equivalence relation[‡] $\sim$. Given the data of

- a set Q, and
- a morphism $q\colon X \to Q$ such that $q(x) = q(x')$ if and only if $x \sim x'$,

we say that these data form a *quotient* of X by $\sim$ if they satisfy the following universal property: for all functions $f\colon X \to Y$ such that $f(x) = f(x')$ if $x \sim x'$, there is a unique function $g\colon Q \to Y$ such that the following diagram commutes:

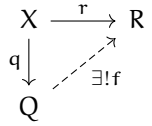


[‡]Recall that an equivalence relation is a reflexive ($\forall x \in X, x \sim x$), symmetric ($\forall x, y \in X,\ x \sim y \implies y \sim x$), and transitive ($\forall x, y, z \in X,\ x \sim y$ and $y \sim z \implies x \sim z$) relation.

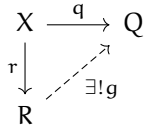
One way to say this is that the quotient morphism q is *universal* or *initial* among all morphisms that respect the equivalence relation $\sim$. As with products, the quotient is unique.

Proposition 2.2.8. *If $q\colon X \to Q$ and $r\colon X \to R$ are two quotients of X by $\sim$, then they are isomorphic.*

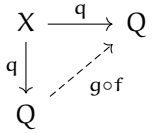
Proof. Since r respects the equivalence relation, we get from the universal property of Q a morphism



Similarly, since q respects the equivalence relation, we get from the universal property of R a morphism



Then $g \circ f$ fills in the diagram



but so does the identity function id_Q, so by uniqueness in the universal property of Q, we see that $g \circ f = \mathrm{id}_Q$. A similar argument shows that $f \circ g = \mathrm{id}_R$, and therefore $Q \cong R$. □

Given a set X with an equivalence relation $\sim$, we can now construct the set Q. This is also often written $X/\sim$.

Definition 2.2.9. Let X be a set with an equivalence relation. The set $X/\sim$ is the set whose elements are equivalence classes of $\sim$. More precisely, for each $x \in X$, let $[x]$ be the set

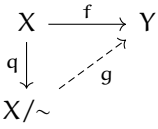
$$[x] := \left\{ x' \in X \;\middle|\; x' \sim x \right\}.$$

This is called the *equivalence class of x under $\sim$*. Then

$$X/\sim := \left\{ [x] \;\middle|\; x \in X \right\}.$$

Proposition 2.2.10. *The set $X/\sim$, together with the map $q\colon X \to X/\sim$ given by $q(x) = [x]$, is a quotient of X by the equivalence relation $\sim$.*

Proof. We check that it satisfies the universal property of a quotient. Let $f\colon X \to Y$ be any other function such that $f(x) = f(x')$ when $x \sim x'$. Then we can define a function $g\colon X/\sim \to Y$ by $g([x]) = f(x)$. This is well-defined because for any $x' \in [x]$, $x' \sim x$, and therefore $f(x) = f(x')$. This also makes the diagram commute



and g is the only possible function that could do so, because to make the diagram commute we must have $g(q(x)) = f(x)$, and $q(x) = [x]$, so g cannot have any other definition. Hence g is unique, so there exists a unique g that makes the diagram commute, and $X/\sim$ is a quotient of X by $\sim$. □

Example 2.2.11. Let $n \in \mathbb{Z}$. Define an equivalence relation on $\mathbb{Z}$ by $x \sim y$ if and only if $x - y$ is a multiple of n. Then the equivalence classes of this relation are the residue classes mod n; there are n of them. These are often written as (for example)

$$k + n\mathbb{Z} := \left\{ x \in \mathbb{Z} \;\middle|\; x \equiv k \pmod{n} \right\}$$

In this example, $\mathbb{Z}/\sim$ is $\mathbb{Z}/n\mathbb{Z}$.

Example 2.2.12. Define an equivalence relation on $\mathbb{R}$ by $x \sim y$ if and only if $x - y \in \mathbb{Z}$. We write the quotient of $\mathbb{R}$ by this equivalence relation as $\mathbb{R}/\mathbb{Z}$. There is one equivalence class for each real number in $[0, 1)$, and the class of 0 is the same as the class of 1. So as a set, this is equivalent to a unit circle S^1. We'll see a topology on this shortly.

Example 2.2.13. Let's return to the Möbius strip example. Define a relation on $X = [0, 1] \times [0, 1]$ by $(0, t) \sim (0, 1 - t)$ for each $t \in [0, 1]$. This identifies the left side of the square with the right side, but only after a flip. The resulting set $X/\sim$ is a Möbius strip. We'll see how to topologize this in a second.

How do we put a topology on any of these examples? We will have to construct quotients in $\mathcal{T}op$.

Definition 2.2.14. Let X be a topological space equipped with an equivalence relation $\sim$. Let Q be the quotient of X by $\sim$. Then the *quotient topology* on Q is given by the following: a set $U \subseteq Q$ is open in Q if and only if $q^{-1}(U)$ is open in X, where $q \colon X \to Q$ is the quotient map.

Exercise 2.2.15. Check that the topology defined above is actually a topology (check that arbitrary unions and finite intersections of open sets are open).

Example 2.2.16. The topology on $\mathbb{R}/\mathbb{Z}$ is equivalent to the subspace topology on S^1 as a subspace of $\mathbb{R}^2$.

Example 2.2.17. The topology on the Möbius strip is exactly what you'd expect: on the surface of the Möbius strip, the basis of open sets is the open balls, and the partial open balls that spill over the edges. If these do not overlap the "cut" in the Möbius strip, then the preimage of these is a single open ball (or partial open ball spilling over the top or bottom). If an open ball overlaps the cut on the Möbius strip, then it is two pieces in the unit square, which is the union of two open sets as a subspace of $\mathbb{R}^2$.

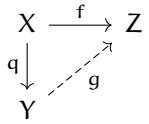
Proposition 2.2.18. *Let X and Y be topological spaces. Let $q \colon X \to Y$ be a surjective continuous map with the property that $U \subseteq Y$ is open if and only if $q^{-1}(Y)$ is open. Then Y is a quotient of X by some equivalence relation $\sim$.*

Proof. There are two ways we could handle this proof. Either we could construct the equivalence relation $\sim$ and show that $Y \cong X/\sim$, which we know has the universal property of a quotient or we could check the universal property directly. We will do the latter.

Define the equivalence relation $\sim$ by $x \sim x'$ if and only if $q(x) = q(x')$. Then let $f \colon X \to Z$ be any continuous function such that $f(x) = f(x')$ when $x \sim x'$. Then we may define a function $g \colon Y \to Z$ by $g(y) = f(x)$ for any $x \in q^{-1}(y)$.

Note that g is well-defined: if $x, x' \in q^{-1}(y)$, then $q(x) = q(x')$ and so $x \sim x'$ and therefore $f(x) = f(x')$. So it doesn't matter which $x \in q^{-1}(y)$ we pick.

Also note that g makes the diagram below commute, by construction.



Finally, g is continuous. Let $W \subseteq Z$ be open. Then we want to show that $g^{-1}(W)$ is open in Y. But since $V \subseteq Y$ is open if and only if $q^{-1}(V)$ is open in X, it suffices to show that $q^{-1}(g^{-1}(W))$ is open in X. But $g \circ q = f$, so

$$q^{-1}(g^{-1}(W)) = f^{-1}(W)$$

and f is continuous so this is open. Hence g is a unique continuous function that makes the diagram commute, and therefore Y satisfies the universal property of the quotient. □

Remark 2.2.19. It is not true that any surjective continuous map $q\colon X \to Y$ is a quotient map in $\mathcal{T}op$, in contrast to the situation in $\mathcal{S}et$ where surjectivity is the only condition necessary to find a quotient map. The difference is that the topology on the target Y must be the finest topology such that q is continuous, and this only happens when $U \subseteq Y$ is open if and only if $q^{-1}(U)$ is open.

It is also not true that a quotient map must be an open map.

2.2.3 Gluing

With the quotient topology, we've learned how to build new topological spaces by identifying parts of them together. But what if we have two different topological spaces, and we want to stick one to another? That's where gluing comes in. The category-theory word for this construction is a pushout.

Definition 2.2.20. In any category $\mathcal{C}$, the *pushout* of a diagram

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ {\scriptstyle g}\downarrow & & \\ Y & & \end{array}$$

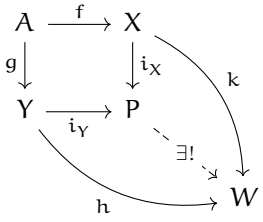
is an object $P \in \mathrm{Ob}(\mathcal{C})$ together with morphisms $i_X\colon X \to P$ and $i_Y\colon Y \to P$ such that the square

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle i_X} \\ Y & \xrightarrow[i_Y]{} & P \end{array}$$

commutes in $\mathcal{C}$ and for any other $W \in \mathrm{Ob}(\mathcal{C})$ with morphisms $k\colon X \to P$ and $h\colon Y \to P$ that make the square

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle k} \\ Y & \xrightarrow[h]{} & W \end{array}$$

commute, there is a unique morphism $P \to W$ such that the diagram



commutes in $\mathcal{C}$.

The notation for the pushout of a diagram $X \leftarrow A \rightarrow Y$ is $X \cup_A Y$. As usual, pushouts are unique up to isomorphism. The proof is much the same as the uniqueness of quotients and the uniqueness of products. We leave the proof as an exercise.

Proposition 2.2.21. *If* P *and* Q *are two pushouts of* $Y \xleftarrow{g} A \xrightarrow{f} X$*, then* $P \cong Q$.

Example 2.2.22. For sets A and B, the pushout of $A \leftarrow \emptyset \rightarrow B$ is the *disjoint union* of A and B. Recall that the disjoint union of two sets A and B is the set $(A \times \{0\}) \cup (B \times \{1\})$, where we add the labels 0 and 1 to separate A and B, ensuring that the sets have an empty intersection. The notation for a disjoint union is $A \sqcup B$ or $A \amalg B$.

Example 2.2.23. For sets A and B, the pushout of $A \hookleftarrow A \cap B \hookrightarrow B$ is $A \cup B$. This is the sense in which we think about the pushout as "gluing" A and B together along their intersection.

We are now ready to construct the pushout in $\mathcal{S}et$.

Proposition 2.2.24. *Let* $X \xleftarrow{f} A \xrightarrow{g} Y$ *be a diagram of sets and functions. Generate an equivalence relation on the disjoint union* $X \amalg Y$ *by* $f(a) \sim g(a)$ *for all* $a \in A$*. Let*

$$X \cup_A Y := (X \sqcup Y)/\sim,$$

equipped with functions $i_X \colon X \to X \cup_A Y$ *and* $i_Y \colon Y \to X \cup_A Y$ *given by* $i_X(x) = [x]$ *and* $i_Y(y) = [y]$*. Then* $X \cup_A Y$ *is the pushout of the diagram* $X \xleftarrow{f} A \xrightarrow{g} Y$*.*

Proof. To show that this has the universal property, let W be any set with morphisms $k \colon X \to P$ and $h \colon Y \to P$ that make the square

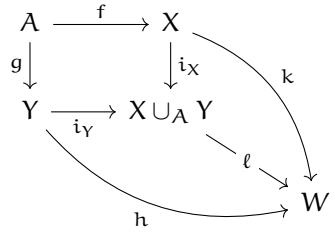
$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle k} \\ Y & \xrightarrow[h]{} & W \end{array}$$

commute. To get a function $X \cup_A Y \to W$, we define

$$\ell(z) = \begin{cases} k(z) & z \in X, \\ h(z) & z \in Y. \end{cases}$$

This is well-defined because if $z = f(a)$ for some $a \in A$, then $\ell(z) = k(f(a))$, and if $z = g(a)$, then $\ell(z) = h(g(a))$, but the square commutes so $k \circ f = h \circ g$, and therefore when we identify $g(a)$ with $f(a)$ in $X \cup_A Y$, we get a the same result applying ℓ to either $f(a)$ or $g(a)$.

Then by construction, $\ell \circ i_X = k$, and $\ell \circ i_Y = h$, so the diagram



commutes.

Finally, ℓ is unique because we must have $\ell \circ i_X = k$ and $\ell \circ i_Y = h$ for the diagram to commute, so there was no other option for defining ℓ in the first place. □

We can also give this a topology.

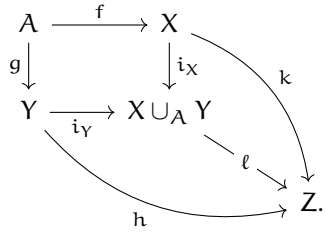
Proposition 2.2.25. *Let $X \xleftarrow{f} A \xrightarrow{g} Y$ be a diagram of topological spaces and continuous functions. The pushout of this diagram in $\mathcal{T}op$ is the set $X \cup_A Y$ from the previous proposition, with the topology given by $U \subseteq X \cup_A Y$ is open if and only if $i_X^{-1}(U) \subseteq X$ and $i_Y^{-1}(U) \subseteq Y$ are open.*

Proof. We check the universal property. The form of the underlying set is already guaranteed by the previous proposition. It is an exercise to prove that the open sets of the form above actually define a topology. Observe that by construction of this topology, i_X and i_Y are guaranteed to be continuous.

Let Z be another topological space and let $k\colon X \to Z$ and $h\colon Y \to Z$ be continuous functions such that the square

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle k} \\ Y & \xrightarrow[h]{} & Z \end{array}$$

commutes. Since $X \cup_A Y$ is the pushout *as sets*, there is a unique function $\ell\colon X \cup_A Y \to W$ that makes the diagram below commute.



It remains to show that ℓ is continuous. Let $V \subseteq Z$ be open. Then we want to show that $\ell^{-1}(V)$ is open in $X \cup_A Y$. This happens if and only if $i_X^{-1}(\ell^{-1}(V))$ and $i_Y^{-1}(\ell^{-1}(V))$ are open, by the definition of the topology on $X \cup_A Y$. But

$$\begin{aligned} i_X^{-1}(\ell^{-1}(V)) &= (h \circ i_X)^{-1}(V) = k^{-1}(V) \\ i_Y^{-1}(\ell^{-1}(V)) &= (h \circ i_Y)^{-1}(V) = h^{-1}(V), \end{aligned}$$

which are both open because k and h are continuous. So by the definition of the topology on $X \cup_A Y$, $\ell^{-1}(V)$ is open, so ℓ is continuous. Uniqueness of ℓ as a continuous function follows from uniqueness of ℓ from the previous proposition. □

Example 2.2.26. The sphere S^2 is the pushout of two disks D^2 along the boundary circle S^1. Similarly, we can make a 3-sphere sphere S^3 as the pushout of two balls D^3 along their boundary circle S^2 (this is really hard to visualize).

Example 2.2.27. Real projective space $\mathbb{RP}^n$ is the space of all lines through the origin in $\mathbb{R}^{n+1}$. More precisely, define an equivalence relation on $\mathbb{R}^{n+1} \setminus \{0\}$ by $\vec{x} \sim \vec{y}$ if and only if $\vec{x} = \lambda\vec{y}$ for some $\lambda \in \mathbb{R}$. Then real projective n-space is the quotient

$$\mathbb{RP}^n := (\mathbb{R}^{n+1} \setminus \{\vec{0}\})/\sim,$$

with the quotient topology.

There's another way to build this space as a pushout. Let's do the $n = 2$ example. $\mathbb{RP}^2$ is the space of lines through the origin in $\mathbb{R}^3$. Any line that is not contained in the $z = 0$ hyperplane passes through a unique point in the upper hemisphere S^2_+ of S^2,

$$S^2_+{}'' = \{(x, y, z) \in S^2 \mid z > 0\}.$$

This space is homeomorphic to $\mathbb{R}^2$. The rest of the lines, those lines in the $z = 0$ hyperplane, form a copy of $\mathbb{RP}^1$, or the lines through the origin in $\mathbb{R}^2$. Hence,

$$\mathbb{RP}^2 \cong \mathbb{R}^2 \cup_{S^1} \mathbb{RP}^1,$$

glued together along the equatorial circle S^1 of S^2 by identifying antipodal points. We can recurse to see that $\mathbb{RP}^1 \cong \mathbb{R}^1 \cup \mathbb{RP}^0 \cong \mathbb{R}^1 \cup \mathbb{R}^0$, where $\mathbb{R}^0$ is glued onto $\mathbb{R}^1$ "at infinity." More generally,

$$\mathbb{RP}^n \cong \mathbb{R}^n \cup \mathbb{R}^{n-1} \cup \cdots \cup \mathbb{R}^1 \cup \mathbb{R}^0.$$

It is an exercise to figure out exactly how these are glued together.

2.3 Hausdorff Spaces/Separating Points

Thus far, we've thought mostly about building spaces from ones that we already know. But this can lead to some pretty wacky and hard-to-understand pictures (imagine doing this in 4D, or 5D, or ...). How do we know if two constructions produce homeomorphic spaces?

Recall the notion of topological properties:

Definition 2.3.1. A *topological property* is a property that is preserved by homeomorphism. More precisely, P is a topological property if:

$$X \text{ has property } P \text{ and } X \cong Y \implies Y \text{ has property } P.$$

Remark 2.3.2. This is a categorical notion, specific to the category $\mathcal{T}op$ of topological spaces and continuous functions. In any category $\mathcal{C}$, we can say that a property is specific to $\mathcal{C}$ whenever it is preserved by isomorphisms in $\mathcal{C}$. In $\mathcal{S}et$, we have set-theoretic properties. In algebraic categories like $\mathcal{A}b$, we have algebraic properties.

This also lets us distinguish spaces that are not homeomorphic by using the converse. If P is a topological property and X has property P but Y does not, then X cannot be homeomorphic to Y.

Example 2.3.3. Cardinality is a topological property, because homeomorphisms are bijections. Therefore, if two spaces have different cardinalities, they cannot be homeomorphic. In particular, a point $\{*\}$ and $\mathbb{R}$ are not homeomorphic.

The topological property we study in this section is called Hausdorff, named after Felix Hausdorff who included this as an axiom in his definition of a topological space (the modern definition does not include this axiom).

Definition 2.3.4. A topological space X *separates points* (or is a *Hausdorff space*) if for every $x, y \in X$, there exists open subsets $U \ni x$ and $V \ni y$ such that U and V are disjoint.

The idea is that in a Hausdorff space, you should be able to zoom in far enough on x or y such that you can't see the other. This is certainly true in Euclidean space, but it's not true in all other spaces. Another reason to think about Hausdorff spaces is that limits of sequences in Hausdorff spaces converge to only one point (yes, it's possible that sequences in non-Hausdorff spaces can have multiple different limits, which is very distressing).

Example 2.3.5. Any metric space separates points. Given a metric space (X, d) and two points $x, y \in X$, then we can build open balls around x and y of radius $d(x, y)/2$ that do not intersect each other.

Example 2.3.6. If X is any set, we can give X the indiscrete topology where the only open sets are $\emptyset$ and X itself. This space is not Hausdorff; any open neighborhood of x is the whole of X itself, so you can't have disjoint neighborhoods of two points.

Example 2.3.7. Let L be the pushout of $\mathbb{R} \hookleftarrow \mathbb{R} \setminus \{0\} \hookrightarrow \mathbb{R}$, in which both maps are inclusions of $\mathbb{R} \setminus \{0\}$ into $\mathbb{R}$; this is the *line with two origins*. More specifically, this space L is constructed as the quotient of $\mathbb{R} \sqcup \mathbb{R}$ by $(x, 0) \sim (x, 1)$ for all $x \neq 0$, but $(0, 0)$ and $(0, 1)$ remain distinct – these are the two origins.

You'll prove on your homework that this space does not separate points.

This last example shows that the Hausdorff property is not preserved by taking pushouts. All three pieces of the diagram $\mathbb{R} \hookleftarrow \mathbb{R} \setminus \{0\} \hookrightarrow \mathbb{R}$ are Hausdorff, but the pushout L is not. Fortunately, many constructions do preserve the Hausdorff property.

Lemma 2.3.8. *The Hausdorff property is preserved by subspaces, products, and coproducts (but not quotients). To be precise:*

(a) If X separates points and A is a subspace of X, then A also separates points.

(b) If X and Y separate points, then so does $X \times Y$.

(c) If X and Y separate points, then so does $X \sqcup Y$.

Proof. (a) Let $a, b \in A$. Then in X, there are disjoint open sets $U \ni a$ and $V \ni b$. Intersecting these with A, we get disjoint open neighborhoods $A \cap U \ni a$ and $A \cap V \ni b$ in A. Hence, A separates points.

(b) Let (x, y) and (x', y') be points in $X \times Y$. Then because X separates points, there are disjoint neighborhoods $U \ni x$ and $U' \ni x'$. Similarly, because Y separates points, there are disjoint neighborhoods $V \ni y$ and $V' \ni y'$. Then $U \times V \ni (x, y)$ and $U' \times V' \ni (x', y')$ are disjoint neighborhoods in the product.

(c) Let $a, b \in X \sqcup Y$. If both $a, b \in X$ or both $a, b \in Y$, then we are done because X and Y both separate points. Otherwise, assume (without loss of generality) that $a \in X$ and $b \in Y$. Take any neighborhoods $U \ni a$ in X and $V \ni b \in Y$ Because this is the disjoint union, we've removed any possibility of intersection between X and Y, so U and V are disjoint because $U \subseteq X$ and $V \subseteq Y$.

□

Proposition 2.3.9. *If X is a Hausdorff space, then any finite subset of X is closed.*

Proof. Because finite unions of closed sets are closed, it suffices to prove that for any $x \in X$, $\{x\}$ is closed. To do so is equivalent to proving that $X \setminus \{x\}$ is open. So let $y \in X \setminus \{x\}$. Because X separates points, there are disjoint open sets $U \ni x$ and $V \ni y$. Since $U \cap V = \emptyset$ and $x \in U$, then $V \subseteq X \setminus \{x\}$. Since y was arbitrary, this shows that any point $y \in X \setminus \{x\}$ has an open neighborhood V_y contained in $X \setminus \{x\}$, so we may write

$$X \setminus \{x\} = \bigcup_{y \in X \setminus \{x\}} V_y,$$

which is a union of open sets. Hence, $X \setminus \{x\}$ is open and so $\{x\}$ is closed. □

Chapter 3

The stuff that's on test 3

3.1 Compact Spaces

We continue our study of topological properties with the notion of compact spaces. In analysis, you may have learned that the compact subspaces of $\mathbb{R}^n$ are the closed and bounded ones. Compact spaces are important because of the extreme value theorem: that a continuous function with a compact domain attains a maximum and a minimum on its domain, but not all spaces are subspaces of $\mathbb{R}^n$. So we need to think of another definition that we can use for a general topological space.

Definition 3.1.1. Let X be a space. An *open cover* of X is a collection $\mathcal{U}$ of open sets

$$\mathcal{U} = \left\{ U_i \subseteq X \;\middle|\; U_i \text{ is open }, i \in I \right\}$$

whose union is all of X:

$$\bigcup_{i \in I} U_i = X.$$

We also make two auxiliary definitions:

- If $J \subseteq I$, then we say that $\{U_j \mid j \in J\}$ is a *subcover* of $\mathcal{U}$.
- If I is finite, then $\{U_i \mid i \in I\}$ is a *finite cover*.

Remark 3.1.2. If A is a subspace of X, and $\mathcal{U} = \{U_i \mid i \in I\}$ is a collection of open subsets of X such that $A \subseteq \bigcup_{i \in I} U_i$, then we say that $\mathcal{U}$ is an *open cover* of A, even though technically the collection $\{U_i \cap A \mid i \in I\}$ is the collection of open sets of A that strictly fits the definition above.

Example 3.1.3. Let $X = \mathbb{R}^2$ with the standard topology. This has an open cover by balls of radius 1 whose centers have integer coordinates. But if we remove any one of these balls $B((a,b),1)$, then it is no longer an open cover, because it no longer covers the point (a,b).

Example 3.1.4. The interval $[0,1]$ has an open cover by sets $\left(\frac{1}{n+2}, \frac{1}{n}\right)$ for $n \in \mathbb{N}$, together with $(-0.1, 0.1)$ and $(0.9, 1.1)$. This open cover actually has a finite subcover, consisting of the sets

$$(-0.1, 0.1), \left(\tfrac{1}{11}, \tfrac{1}{9}\right), \left(\tfrac{1}{10}, \tfrac{1}{8}\right), \ldots, \left(\tfrac{1}{3}, 1\right), (0.9, 1.1).$$

It turns out that this property of having a finite subcover will characterize compactness.

Definition 3.1.5. A topological space X is *compact* if every open cover has a finite subcover.

Example 3.1.6. By this definition, $\mathbb{R}^2$ is not compact, because we exhibited an example of an open cover with no finite subcover.

3.1.1 Closed Intervals are Compact

It takes a lot more work to prove that our second example $[0,1]$ is compact.

Lemma 3.1.7 (Topological Induction)**.** *Let C be a subspace of $[0,1]$. Assume that:*

(a) $0 \in C$

(b) If $b \in C$ and $0 \leq a \leq b$, then $a \in C$.

(c) If $b \in C$ and $b < 1$, then there exists some $\varepsilon > 0$ such that $b + \varepsilon \in C$

(d) The supremum[*]

Then $C = [0,1]$.

This lemma works equally well with $[0,1]$ replaced by an arbitrary interval $[a,b]$.

Proof. Let C be as in the statement of the lemma. We want to show that $C = [0,1]$.

By (a), C is nonempty. Furthermore, $C \subseteq [0,1]$, so C is bounded above by 1. The least upper bound property of $\mathbb{R}$ guarantees that the supremum of C exists and moreover, $\sup(C) \leq 1$.

By (d), $\sup(C) \in C$, and by (b), every number between 0 and $\sup(C)$ is in C as well. Hence, $C = [0, \sup(C)]$.

So it remains to show that $\sup(C) = 1$. If $\sup(C) < 1$, then (c) would guarantee that $\sup(C) + \varepsilon \in C$, contradicting the fact that $\sup(C)$ must be an upper bound of C. Therefore, $\sup(C) = 1$. □

Theorem 3.1.8. *The subspace $[0,1] \subseteq \mathbb{R}$ is compact.*

Proof. We must show that every open cover of $[0,1]$ has a finite subcover. To that end, let $\mathcal{U} = \{U_i \mid i \in I\}$ be an open cover of $[0,1]$. Let

$$X = \left\{ x \in [0,1] \;\middle|\; [0,x] \text{ is covered by the union of finitely many } U_i\text{s} \right\} \subseteq [0,1].$$

The idea is to apply topological induction to X to show that $X = [0,1]$, whence we are done.

To that end, we check the four conditions of the topological induction lemma:

(a) We want to show that $0 \in X$. But since $\mathcal{U}$ is an open cover of $[0,1]$, then $0 \in U_i$ for some i. Hence, $\{0\} = [0,0] \subseteq U_i$, so $[0,0]$ is covered by a single open set in $\mathcal{U}$, so $0 \in X$.

(b) If $b \in X$, then $[0,b]$ has a finite subcover. If $0 \leq a \leq b$, then $[0,a] \subseteq [0,b]$, and it must also have a finite subcover (the same finite subcover as the one that covers $[0,b]$ works). Hence, $a \in X$.

(c) Let $b \in X$. Then $[0,b]$ has a finite subcover, say by open sets $U_0, \ldots, U_n \in \mathcal{U}$. Say that $b \in U_n$. Since U_n is open, there is some $\delta > 0$ such that $B(b,\delta) \subseteq U_n$. Then $b + \delta/2 \in U_n$, and in particular $\{U_0, \ldots, U_n\}$ is still an open cover of $[0, b+\delta/2]$. Set $\varepsilon = \delta/2$.

(d) Finally, we have to check that the supremum of X is in X. Let $s = \sup(X)$. By the previous three properties, we know that $X = [0,s]$ or $X = [0,s)$. We want to rule out the latter case.

Since $\mathcal{U}$ is an open cover, there is some i such that $s \in V$ for some $V \in \mathcal{U}$. By openness of V, there is some $\delta > 0$ such that $B(s,\delta) \subseteq V$. In particular, $(s-\varepsilon, s] \subseteq V$, where $\varepsilon = \delta/2$. Since $s - \varepsilon \in X$, we know that $[0, s-\varepsilon]$ has a finite subcover. Adding V to this finite subcover preserves its finiteness, while covering all of $[0,s] = [0, s-\varepsilon] \cup (s-\varepsilon, s]$. Hence, $s \in X$.

Therefore, we can apply topological induction to say that $X = [0,1]$. Since X has a finite subcover, then so to $[0,1]$ has a finite subcover. Hence, $[0,1]$ is compact. □

[*]The *supremum* of $X \subseteq \mathbb{R}$ is the least upper bound of X. We say that $L \in \mathbb{R}$ is an *upper bound* for X if $x \leq L$ for all $x \in X$. Then $\sup(X) \in \mathbb{R}$ is the *supremum* of X if $\sup(X)$ is an upper bound for X and for all other upper bounds L, $\sup(X) \leq L$. For example, the supremum of (a,b) is b, and the supremum of $[a,b]$ is also b, but $[0,\infty)$ has no upper bound and therefore no supremum.

3.1.2 Properties of Compact Spaces

Since it's so hard to prove that spaces are compact, we investigate ways in which we can deduce compactness from knowing that some other spaces are compact.

Proposition 3.1.9. *Let $f\colon X \to Y$ be a continuous function. If X is compact, then $f(X) \subseteq Y$ is compact.*

Proof. Without loss of generality, assume that f is a surjective function, so that $f(X) = Y$. We want to show that Y is compact. More precisely, we want to show that every open cover of Y has a finite subcover.

Let $\mathcal{U} = \{U_i \mid i \in I\}$ be an open cover of Y. Then we can build an open cover of X by taking

$$\mathcal{V} = \{f^{-1}(U_i) \mid i \in I\}.$$

Since f is continuous, each $f^{-1}(U_i)$ is open, and these cover X because for each $x \in X$, $f(x) \in U_i$ for some i since the U_is cover Y.

Then because X is compact, this open cover $\mathcal{V}$ has a finite subcover, say

$$X = f^{-1}(U_1) \cup f^{-1}(U_2) \cup \ldots \cup f^{-1}(U_n).$$

Let's apply f to both sides of this equation. Since f is surjective, we get $f(X) = Y$ on the left. On the right, we can again take advantage of the fact that f is surjective to get $f(f^{-1}(U_i)) = U_i$. Therefore,

$$Y = U_1 \cup U_2 \cup \ldots \cup U_n.$$

This is a finite subcover of $\mathcal{U}$ that covers Y, so Y is compact. □

Since a homeomorphism is (among other things) a continuous surjection, we learn that:

Corollary 3.1.10. *Compactness is a topological property.*

Proposition 3.1.11. *Let X be a compact space. Let K be a closed subspace of X. Then K is compact.*

Proof. Let $\mathcal{U} = \{U_i \subseteq K \mid i \in I\}$ be an open cover of K. Since K has the subspace topology, we can write $U_i = V_i \cap K$ for some open set $V_i \subseteq X$. These V_i may not cover X, but they do cover K. Make an open cover of X by taking all of the V_is and also the open set $X \setminus K$:

$$\mathcal{V} = \left\{V_i \subseteq X \,\middle|\, i \in I\right\} \cup \left\{X \setminus K\right\}.$$

Then $\mathcal{V}$ is an open cover of X, because the V_is cover K and $X \setminus K$ covers everything outside of K. Hence, it has a finite subcover, say

$$X = V_1 \cup V_2 \cup \ldots \cup V_n \cup (X \setminus K).$$

Then intersecting this finite subcover with K, we get

$$K = U_1 \cup U_2 \cup \ldots \cup U_n,$$

which is a finite subcover of $\mathcal{U}$ covering K. Hence, K is compact. □

It is not always true that compact subspaces are closed, but it is true if we add the Hausdorff hypothesis.

Theorem 3.1.12. *Let X be a Hausdorff space, and let $A \subseteq X$ be compact. Then A is closed.*

Proof. Proving that A is closed is equivalent to proving that $X \setminus A$ is open. We will prove that $X \setminus A$ is open by proving that for each $x \in X \setminus A$, there is some open neighborhood $U \ni x$ such that $U \subseteq X \setminus A$.

Fix $x \in X \setminus A$. Because X is Hausdorff, for each $a \in A$, there are open sets $U_a \ni x$ and $V_a \ni a$ such that $U_a \cap V_a = \emptyset$. We use the notation U_a and V_a to emphasize the dependence of these open sets on a,

even though $x \in U_a$. Since x is fixed, changing the point a will result in varying the open set containing x, depending on a.

The collection $\{V_a \mid a \in A\}$ is an open cover of A. Since A is compact, this means that there is a finite subcover of this set, say

$$A \subseteq V_{a_1} \cup V_{a_2} \cup \ldots \cup V_{a_n}$$

for some finite number of points $a_1, \ldots, a_n \in A$. Let

$$V = V_{a_1} \cup V_{a_2} \cup \ldots \cup V_{a_n}$$

and let

$$U = U_{a_1} \cap U_{a_2} \cap \ldots \cap U_{a_n}.$$

Then by construction, $A \subseteq V$. We also know that V and U are disjoint, because $V_{a_i} \cap U_{a_i} = \emptyset$ for all i, and $U \subset U_{a_i}$ for all i. Since V and U are disjoint, then A and U are disjoint since $A \subseteq V$. Furthermore, U is open because it is an intersection of finitely many open sets. Finally, $x \in U$ since $x \in U_{a_i}$ for all i.

This set U is the open neighborhood of x that we wanted to find. We know that $x \in U$, and $U \cap A = \emptyset$, so $U \subseteq X \setminus A$. Therefore, $X \setminus A$ is open, so A is closed. □

We can use this to show that compact subspaces of $\mathbb{R}^n$ are closed and bounded. Recall that a subset $A \subseteq \mathbb{R}^n$ is *bounded* if there is some $r > 0$ such that $A \subseteq B(\vec{0}, r)$.

Corollary 3.1.13. *Let $A \subseteq \mathbb{R}^n$ be a compact subspace. Then A is closed and bounded.*

Proof. By Theorem 3.1.12, A must be a closed subspace of $\mathbb{R}^n$. Second, consider the open cover of $\mathbb{R}^n$ consisting of balls of increasing radius centered at the origin.

$$\left\{ B(\vec{0}, r) \;\middle|\; r = 1, 2, 3, \ldots \right\}$$

This is an open cover of $\mathbb{R}^n$ (the union of all such balls is all of $\mathbb{R}^n$), and therefore the collection

$$\left\{ A \cap B(\vec{0}, r) \;\middle|\; r = 1, 2, 3 \ldots \right\}$$

is an open cover of A. Since A is compact, this open cover must have a finite subcover, and therefore $A \subseteq B(\vec{0}, r)$ for some radius r. In other words, A is bounded. □

We want to also prove the converse of this: that closed and bounded subspaces of $\mathbb{R}^n$ are compact. To do so, however, we will first need to analyze when a product space is compact. We will come back to this in the next section.

Finally, we have a useful theorem about when a continuous bijection is a homeomorphism.

Theorem 3.1.14 (Fundamental Theorem of Topology)**.** *Let $f \colon X \to Y$ be a continuous function that is also a bijection. Assume that X is compact and that Y is Hausdorff. Then f is a homeomorphism.*

Proof. Remember from Proposition 1.5.22 that a homeomorphism is equivalently a continuous bijection that sends open sets to open sets. So it remains to show that for any $U \subseteq X$ open, $f(U)$ is open in Y. Or equivalently, that for any $K \subseteq X$ closed, that $f(K)$ is closed in Y (think $K = X \setminus U$).

To that end, let $K \subseteq X$ be closed. Then because X is compact, K is compact as well by Proposition 3.1.11. Therefore, by Proposition 3.2.8, $f(K)$ is compact. Finally, by Theorem 3.1.12, $f(K)$ is closed. Hence, f sends closed sets to closed sets, and therefore f is a homeomorphism. □

3.1.3 Inheritance of Compactness

Whenever we have a new topological property, we should ask how it behaves with respect to the constructions from chapter 2. In particular, we wonder how this topological property behaves with respect to subspaces, quotients, products, and coproducts. (Since pushouts are built out of coproducts and quotients, this will also give us the answer for pushouts.)

For compactness, we already answered the question of subspaces in Proposition 3.1.11: the answer is that closed subspaces of compact spaces are compact, but not necessarily open ones.

Example 3.1.15. $[0,1]$ is a compact space, but $(0,1)$ is not compact because it is homeomorphic to $\mathbb{R}$, which is not compact.

We also already have the tools we need to figure out whether or not quotients of compact spaces are compact.

Proposition 3.1.16. *If Q is a quotient of X by some equivalence relation, and X is compact, then Q is compact.*

Proof. Let $q\colon X \to Q$ be the quotient map. Since X is compact and q is surjective, Proposition 3.2.8 shows that Q is compact as well. □

The tricky part for compactness is to deal with products. Coproducts are not nearly as hard as products, so you'll tackle that for your homework.

Lemma 3.1.17 (Tube Lemma)**.** *Let X and Y be spaces such that Y is compact. Let $x \in X$. Suppose that $W \subseteq X \times Y$ is an open neighborhood[†] of $\{x\} \times Y$. Then there is an open neighborhood $U \ni x$ in X such that $U \times Y \subseteq W$.*

Proof. Since W is open, and $W \supseteq \{x\} \times Y$, then for any $y \in Y$ there are open sets $x \in U_y \subseteq X$ and $y \in V_y \subseteq Y$ such that $U_y \times V_y \subseteq Y$. The collection

$$\left\{ V_y \;\middle|\; y \in Y \right\}$$

is then an open cover of Y. Since Y is compact, there is a finite subcover, say

$$Y = V_{y_1} \cup V_{y_2} \cup \ldots \cup V_{y_n}.$$

Let

$$U = U_{y_1} \cap U_{y_2} \cap \ldots \cap U_{y_n}.$$

Since $x \in U_y$ for all $y \in Y$, then $x \in U$. Also, U is open because it is a finite intersection of open sets. We claim that U is the set we desire.

$$U \times Y \subseteq U \times (V_{y_1} \cup V_{y_2} \cup \ldots \cup V_{y_n}) \subseteq (U_{y_1} \times V_{y_1}) \cup (U_{y_2} \times V_{y_2}) \cup \ldots \cup (U_{y_n} \times V_{y_n}) \subseteq W$$

□

Theorem 3.1.18. *Let X and Y be compact spaces. Then $X \times Y$ is compact.*

Proof. To prove that $X \times Y$ is compact, we have to prove that every open cover has a finite subcover.

Let $\mathcal{A} = \{W_i \mid i \in I\}$ be an open cover of $X \times Y$. For each $x \in X$, the subspace $\{x\} \times Y$ is homeomorphic to Y. Therefore, $\{x\} \times Y$ is compact and may therefore be covered by finitely many elements of $\mathcal{A}$, say

$$\{x\} \times Y \subseteq W_1 \cup W_2 \cup \ldots \cup W_n.$$

[†]If X is a space and $A \subseteq X$ is a subset of X, then we say that an open set $U \subseteq X$ is an *open neighborhood* of A if $U \supseteq A$. This generalizes the idea of an open neighborhood of a point to an open neighborhood of a set.

Then $W = W_1 \cup W_2 \cup \ldots \cup W_n$ is an open set containing $\{x\} \times Y$, so there is an open neighborhood $U_x \ni x$ in X such that $U_x \times Y \subseteq W$. In particular,

$$U_x \times Y \subseteq W = W_1 \cup W_2 \cup \ldots \cup W_n.$$

In other words, $U_x \times Y$ is covered by finitely many open sets $W_1, \ldots, W_n$ from $\mathcal{A}$.

Now we have to use compactness of X. In the previous paragraph, we constructed an open set $U_x \ni x$ for each $x \in X$; these form an open cover of X. Since X is compact, there is a finite number of these that covers X, say

$$X = U_1 \cup U_2 \cup \ldots \cup U_n.$$

Then the union of the tubes $U_i \times Y$ is all of $X \times Y$.

$$X \times Y = \bigcup_{i=1}^{n} U_i \times Y.$$

Since each $U_i \times Y$ may be covered by a finite number of open sets from $\mathcal{A}$, and there are finitely many of them, so too may $X \times Y$ be covered by a finite number of open sets from $\mathcal{A}$. □

Example 3.1.19. This theorem shows that the closed square $[0,1] \times [0,1]$ is compact, because $[0,1]$ is compact. By induction, we can also show that the closed box $[0,1]^n \subseteq \mathbb{R}^n$ is compact.

3.1.4 Compact Subspaces of $\mathbb{R}^n$

We have already seen in Corollary 3.1.13 that a compact subspace of $\mathbb{R}^n$ is both closed and bounded. Here, we prove the converse as well.

Theorem 3.1.20 (Heine–Borel)**.** *A subset $A \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Proof. If A is compact, then it is closed and bounded by Corollary 3.1.13.

Conversely, assume that A is both closed and bounded. Then $A \subseteq B(\vec{0}, r)$ for some radius $r > 0$. Enlarging the ball a little bit, we also see that $A \subseteq [-r, r]^n$. In fact, A will be a closed subspace of $[-r, r]^n$, so by Proposition 3.1.11, it suffices to show that $[-r, r]^n$ is closed. But we know by Theorem 3.1.18 that products of compact spaces are compact, and we know that $[-r, r]$ is compact (because it is homeomorphic to $[0, 1]$), so $[-r, r]^n$ is compact. Therefore, A is compact as well. □

3.2 Connected Spaces

Definition 3.2.1. Let X be a topological space.

(a) A *separation* of X is a pair of nonempty open subsets A and B such that $A \cup B = X$ and $A \cap B = \emptyset$.

(b) We say that X is *connected* if there is no separation of X.

If there is a separation $X = A \cup B$ of X, then $X \cong A \sqcup B$. In many ways, the study of connected spaces is the study of disjoint unions.

Example 3.2.2. The space $X = [-1, 0) \cup (0, 1]$ is not connected; $A = [-1, 0)$ and $B = (0, 1]$ is a separation of X.

Example 3.2.3. We can write the space $X = [-1, 1]$ as $[-1, 1] = [-1, 0] \cup (0, 1]$. Setting $A = [-1, 0]$ and $B = (0, 1]$, we have $A \cup B = X$ and $A \cap B = \emptyset$. But this is not a separation because $A = [-1, 0]$ is not closed in $X = [-1, 1]$.

Example 3.2.4. Any space X with the indiscrete topology (the only open sets are X and $\emptyset$) is connected.

Example 3.2.5. The rational numbers $\mathbb{Q}$ are *not* connected. If a is any rational number, then $A = (-\infty, a) \cap \mathbb{Q}$ and $B = (a, \infty) \cap \mathbb{Q}$ form a separation of $\mathbb{Q}$. In fact, the only connected subspaces of $\mathbb{Q}$ are singletons. We call any such space *totally disconnected*.

Example 3.2.6. Consider the subspace of $\mathbb{R}^2$ given by

$$X = \left\{(x, 0) \mid x \in \mathbb{R}\right\} \cup \left\{ \left(x, \tfrac{1}{x}\right) \;\middle|\; x > 0 \right\}$$

If we let A be the x-axis and let B be the graph of the function $y = 1/x$, then these two subsets form a separation of X. Therefore, X is disconnected, even though the two sets get asymptotically close.

The simplest way to tell whether or not a space is connected is to draw it and look. But some spaces are very hard to draw, so we need other ways to tell.

Proposition 3.2.7. *A space X is connected if and only if the only subsets of X which are both open and closed are $\emptyset$ and X.*

Proof. We prove both directions of this proposition by proving the contrapositive, which is the statement:

> A space X is disconnected if and only if there is a proper nonempty subset $A \subseteq X$ such that A is both open and closed.

Assume that X is disconnected. Then there are disjoint, nonempty open sets A and B such that $X = A \cup B$. Since B is open, and $X \setminus B = A$, then A is closed. But A is also open. Since $A \neq \emptyset$ and $A \neq X$, then A is a proper nonempty subset that is both open and closed.

Conversely, assume that $A \subseteq X$ is a proper nonempty subset that is both open and closed. Then $B := X \setminus A$ is open, and $X = A \cup B$ is a separation of X. □

Proposition 3.2.8. *Let $f\colon X \to Y$ be a continuous function with X a connected space. Then $f(X) \subseteq Y$ is connected.*

Proof. Assume without loss of generality that $f(X) = Y$ (so that f is surjective). Assume that $A \subseteq Y$ is both open and closed. If Y is to be connected, we must prove that $A = \emptyset$ or $A = Y$.

Since f is continuous and A is open, then $f^{-1}(A)$ is open. Similarly, since f is continuous and A is closed, then $f^{-1}(A)$ is closed. Hence, $f^{-1}(A) \subseteq X$ is both open and closed. Since X is connected then either $f^{-1}(A) = \emptyset$, or $f^{-1}(A) = X$.

- If $f^{-1}(A) = \emptyset$, then $A = \emptyset$ because f is surjective.
- If $f^{-1}(A) = X$, then $A = Y$ since f is surjective.

Therefore, $A = \emptyset$ or $A = Y$, so Y must be connected. □

Corollary 3.2.9. *Connectedness is a topological property.*

3.2.1 Inheritance of Connectedness

Thus far, we have not yet proved that any spaces are connected, but you can probably guess that $\mathbb{R}$, $[0, 1]$, $(0, 1)$ and many other spaces are connected. We will prove this later, when we get to path-connectedness. For now, let's assume that these spaces are connected and think about how we might build new connected spaces from old ones. As always, we focus on products, coproducts, quotients, and subspaces. For connected spaces, we know that $A \sqcup B$ is always disconnected, so we can skip thinking about coproducts in this context.

Let's start with subspaces. Since $\mathbb{Q} \subseteq \mathbb{R}$ is (totally) disconnected, then it can't be the case that subspaces of a connected space are connected. But we can nevertheless say something about connected subspaces of disconnected spaces.

Proposition 3.2.10. *Let X be a topological space with a separation $X = A \cup B$. Let $Y \subseteq X$ be a connected subspace. Then $Y \subseteq A$ or $Y \subseteq B$.*

We might equally well phrase this proposition as follows: If Y is connected, and $Y \hookrightarrow A \sqcup B$, then $Y \hookrightarrow A$ or $Y \hookrightarrow B$.

Proof. Since A and B are both open in X, then $A \cap Y$ and $B \cap Y$ are both open in Y. These two sets are disjoint (since A and B are disjoint) and their union is Y (since $A \cup B = X$). If they were both nonempty, then they would constitute a separation of Y, so one or the other must be empty. If $A \cap Y = \emptyset$, then $B \cap Y = Y$ and so $Y \subseteq B$. Similarly, if $B \cap Y = \emptyset$, then $Y \subseteq A$. □

As with compact spaces, the quotient of a connected space is connected. This is another corollary of Proposition 3.2.8, but we will state it as a separate proposition.

Proposition 3.2.11. *If X is a connected space, then any quotient Q of X is connected.*

Finally, this brings us to products of connected spaces. This is the hard one, even though it's not quite as hard as products of compact spaces. First, we need a lemma.

Lemma 3.2.12. *Let X be a topological space and let $x \in X$. Let $\{A_i \subseteq X \mid i \in I\}$ be a collection of connected subspaces of X such that $x \in A_i$ for all $i \in I$. Then $\bigcup_{i \in I} A_i$ is connected.*

Proof. Let $Y = \bigcup_{i \in I} A_i$. We prove that Y is connected by contradiction.

Assume that Y is disconnected, and that $Y = B \cup C$ is a separation of Y. Then B and C are nonempty, and $B \cup C = Y$ and $B \cap C = \emptyset$. Since $x \in Y$, then x is in only one of B or C. Without loss of generality, say $x \in C$. Then since $x \in A_i$ for all $i \in I$, and A_i is connected, $A_i \subseteq C$ by Proposition 3.2.10. Since $A_i \subseteq C$ for all i, and Y is the union of the A_i, then $Y \subseteq C$. This contradicts the fact that B was assumed to be nonempty. □

Theorem 3.2.13. *If X and Y are connected spaces, then $X \times Y$ is connected.*

Proof. Let $a \in X$ and $b \in Y$. Then $(a, b) \in X \times Y$, and $X \times \{b\}$ is connected, being homeomorphic to X, and for each $x \in X$, $\{x\} \times Y$ is connected, being homeomorphic to Y. Therefore, the T-shaped subspace

$$T_x = \Big(\{x\} \times Y\Big) \cup \Big(X \times \{b\}\Big)$$

is connected by Lemma 3.2.12. Note that the "cross" $X \times \{b\}$ of the T is always at the same height, but the "stem" $\{x\} \times Y$ of the T moves left-to-right.

Since $(a, b) \in T_x$ for all $x \in X$, then the union of these sets is connected, again by Lemma 3.2.12. But

$$X \times Y = \bigcup_{x \in X} T_x,$$

so $X \times Y$ is connected. □

3.2.2 Connectedness of $[0, 1]$

Theorem 3.2.14. *The unit interval $[0, 1]$ is connected.*

Proof. We again prove this via topological induction (Lemma 3.1.7).

Let $U \subseteq [0, 1]$ be a nonempty open subset of $[0, 1]$ that is both open and closed. To prove that $[0, 1]$ is connected, it suffices to show that $U = [0, 1]$. Since U is nonempty, either $0 \in U$ or $0 \in [0, 1] \setminus U$. Replacing U by its complement (which will also be both open and closed) if necessary, we may assume that $0 \in U$. Let

$$C = \Big\{t \in [0, 1] \mid [0, t] \subseteq U\Big\}.$$

We will prove by topological induction that $C = [0, 1]$, whence $U = [0, 1]$.

(a) $0 \in C$, since $0 \in U$.

(b) Let $b \in C$ and let $0 \leq a \leq b$. To show that $a \in C$, consider that $[0, a] \subseteq [0, b] \subseteq U$, so $[0, a] \subseteq U$. Hence, $a \in C$.

(c) Let $b \in C$ with $b < 1$. We want to show that $b + \varepsilon \in C$ for some $\varepsilon > 0$. To that end, note that $[0, b] \subseteq U$, so $b \in U$. Since U is open, there is some $\delta > 0$ such that $B(b, \delta) \subseteq U$. Set $\varepsilon = \delta/2$. Then $b + \varepsilon \in B(b, \delta) \subseteq U$, and therefore $[0, b + \varepsilon] \subseteq U$ (also using item (b)).

(d) Finally, we must check that the supremum of C is in C[‡]. Since C is nonempty and bounded above, the supremum of C exists. Let $s = \sup(C)$. If $s \notin C$, then $s \notin U$. But since s is the least upper bound of elements in U, then $[0, s) \subseteq U$.

Since U is closed, then the complement $V = [0, 1] \setminus U$ is open. Since $s \notin U$, then $s \in V$. Since V is open, there is some $\delta > 0$ and an open ball $B(s, \delta)$ around s such that $B(s, \delta) \subseteq V$. Set $\varepsilon = \delta/2$. Then $s - \varepsilon \in B(s, \delta) \subseteq V$. Therefore, $s - \varepsilon \in V$ and by definition, $s - \varepsilon \notin U$. This contradicts the fact that s was the supremum of C, and therefore the assumption that $s \notin C$ cannot be.

Hence, $s \in C$, so C contains its supremum.

By topological induction, $C = [0, 1]$, and therefore $U = [0, 1]$ as well. □

Corollary 3.2.15. *For any $a, b \in \mathbb{R}$, $[a, b]$ is connected.*

Proof. $[a, b]$ is homeomorphic to $[0, 1]$, which is connected. □

Corollary 3.2.16. *$\mathbb{R}$ is connected.*

Proof. Suppose for the sake of contradiction that $\mathbb{R}$ is disconnected. That is, assume $\mathbb{R} = A \cup B$, with $A \cap B = \emptyset$ and both A, B nonempty and open in $\mathbb{R}$. Then let $a \in A$ and $b \in B$. Then

$$A_0 = A \cap [a, b] \text{ and } B_0 = B \cap [a, b]$$

constitutes a separation of $[a, b]$, since $a \in A_0$ and $b \in B_0$ (so neither is empty), and $A_0 \cup B_0 = [a, b]$ and $A_0 \cap B_0 = \emptyset$. This contradicts the fact that $[a, b]$ is connected. Therefore, the assumption that $\mathbb{R}$ is disconnected cannot be. □

Remark 3.2.17. This proof actually works to show that any *convex* (see Example 3.3.5) subset of $\mathbb{R}$ is connected. But we can also deduce that from path-connectedness.

3.2.3 Connected Components

Let X be a topological space. Define an equivalence relation $\sim$ on X by $x \sim y$ if and only if there is a connected subspace $A \subseteq X$ such that $x \in A$ and $y \in A$.

Definition 3.2.18. The equivalence classes of X with respect to this equivalence relation are called the *connected components* of X.

Example 3.2.19. If X is connected, it only has one connected component, namely X itself.

Example 3.2.20. If $X = A \sqcup B$ with A and B connected, then there are two connected components: A and B.

Example 3.2.21. The connected components of $\mathbb{Q}$ are the individual rational numbers: each $\{a\}$ is a connected component of $\mathbb{Q}$. This is what it means to say that $\mathbb{Q}$ is *totally disconnected*.

Remark 3.2.22. A space X is connected if and only if X has only one connected component.

[‡]If you use the analysis definition of closed (a set is closed if and only if it contains all its limit points), then you can prove this using limit points. The supremum of C is a limit point of C, and therefore a limit point of U. Since U is closed, it contains all of its limit points. Therefore, the supremum is in U, and hence the supremum is in C.

3.3 Path Connectedness

Thus far we have yet to prove that anything is actually connected. In general, proving that things are connected is tough, whereas proving that things are not connected is more straightforward. Instead of directly proving that anything is connected, we introduce a new topological property: path-connectedness. We will show that path-connectedness implies connectedness, and therefore deduce that many spaces are connected.

Definition 3.3.1. Let X be a topological space. A *path* in X is a continuous function $\gamma\colon [0,1] \to X$.

We say that $\gamma(0) \in X$ is the start of the path, and $\gamma(1) \in X$ is the end of the path.

Definition 3.3.2. A space X is *path-connected* if for every pair of points $x, y \in X$, there is a path $\gamma\colon [0,1] \to X$ such that $\gamma(0) = x$ and $\gamma(1) = y$.

Example 3.3.3. The space $[0,1]$ is itself path-connected. For any $a, b \in \mathbb{R}$, the line segment

$$\gamma(t) = ta + (1-t)b$$

is a path with $\gamma(0) = a$ and $\gamma(1) = b$.

Example 3.3.4. Similarly, $\mathbb{R}^n$ is path-connected. For any $\vec{v}, \vec{u} \in \mathbb{R}^n$, the line segment

$$\gamma(t) = t\vec{v} + (1-t)\vec{u}$$

connects them, and satisfies $\gamma(0) = \vec{v}$ and $\gamma(1) = \vec{u}$.

Example 3.3.5. A subset $A \subseteq \mathbb{R}^n$ is *convex* if for any two points $\vec{a}, \vec{b} \in A$, the line segment connecting $\vec{a}$ and $\vec{b}$ is entirely contained inside A:

$$\left\{ t\vec{a} + (1-t)\vec{b} \;\middle|\; t \in [0,1] \right\} \subseteq A.$$

Any convex subset of $\mathbb{R}^n$ is path-connected.

Proposition 3.3.6. *If $f\colon X \to Y$ is a continuous function, and X is path-connected, then so is $f(X) \subseteq Y$.*

Proof. Without loss of generality, we may assume that $f(X) = Y$ (so that f is surjective). Let $y_0, y_1 \in Y$. We want to find a continuous function $\gamma\colon [0,1] \to Y$ such that $\gamma(0) = y_0$ and $\gamma(1) = y_1$. Since f is surjective, choose $x_0 \in X$ such that $f(x_0) = y_0$, and $x_1 \in X$ such that $f(x_1) = y_1$. Then since X is path connected, there is a path $\gamma\colon [0,1] \to X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$. Then $f \circ \gamma\colon [0,1] \to Y$ is the desired path connecting y_0 and y_1. □

Corollary 3.3.7. *Path-connectedness is a topological property.*

This also tells us that quotients of path-connected spaces are path-connected.

Corollary 3.3.8. *If Q is a quotient of X, and X is path-connected, then so is Q.*

While we're here, we can also show that products of path-connected spaces are path-connected. This is much easier than proving that products of compact spaces are compact, or products of connected spaces are connected.

Proposition 3.3.9. *Let X and Y be path-connected spaces. Then $X \times Y$ is path-connected.*

Proof. Let X and Y be path connected spaces. For any two points (x_0, y_0) and (x_1, y_1) in $X \times Y$, we must find a path starting at one and ending at the other. Since X is connected, we can find a path connecting any two points in X. In particular, we can find a path $\gamma_X\colon [0,1] \to X$ with $\gamma(0) = x_0$ and $\gamma(1) = x_1$. Similarly, since

Y is path connected, we can find a path $\gamma_Y\colon [0,1] \to Y$ such that $\gamma_Y(0) = y_0$ and $\gamma_Y(1) = y_1$. Then we can build a path $\Gamma\colon [0,1] \to X \times Y$ by

$$\Gamma(t) = (\gamma_X(t), \gamma_Y(t)).$$

We have $\Gamma(0) = (\gamma_X(0), \gamma_Y(0)) = (x_0, y_0)$, and $\Gamma(1) = (\gamma_X(1), \gamma_Y(1)) = (x_1, y_1)$, and Γ is continuous because both γ_X and γ_Y are continuous. This is the desired path. Since this works for any pair of points in $X \times Y$, we have shown that $X \times Y$ is path-connected. □

Theorem 3.3.10. *If X is path connected, then X is connected.*

Proof. Assume that X is path-connected. Then suppose for the sake of contradiction that X is not connected. In other words, assume that there is a nonempty, proper subset $A \subseteq X$ which is both open and closed.

Then let $a \in A$ and let $b \in X \setminus A$, and join them by a path $\gamma\colon [0,1] \to X$ such that $\gamma(0) = a$ and $\gamma(1) = b$. Then $\gamma^{-1}(A)$ is a nonempty (because $\gamma(0) \in A$) and proper (because $\gamma(1) \notin A$) subset of $[0,1]$, which by continuity, must be both open and closed in $[0,1]$. This contradicts the connectedness of $[0,1]$. Therefore, X must be connected. □

Finally, we have an example of a space which is connected but not path-connected. Yuck.

Example 3.3.11 (The "Topologist's Sine Curve", [Arm83, Pages 62-63])**.** Let $X \subseteq \mathbb{R}^2$ be the subspace of $\mathbb{R}^2$ defined by the union of

$$Y = \{0\} \times [0,1] \quad \text{and} \quad Z = \left\{ \left(x, \sin\left(\tfrac{\pi}{x}\right)\right) \;\middle|\; 0 < x \leq 1 \right\}.$$

This space is connected; each piece Z and Y in the union is connected, so the only possibility for a separation is "between" the two components. But any open set containing $Y = \{0\} \times [0,1]$ intersects Z nontrivially, so this is impossible. (This isn't the most rigorous, but this example is terrible so we're not going to linger on it.)

On the other hand, this space isn't path connected. To see this, let $\gamma\colon [0,1] \to X$ be a path such that $\gamma(0) \in Y$ and $\gamma(1) \in Z$. We will show that this is impossible.

Since Y is closed in $\mathbb{R}^2$, it is a closed subset of X, and therefore $\gamma^{-1}(Y) \subseteq [0,1]$ is a closed subset. Note also that $0 \in \gamma^{-1}(Y)$, so if we can show that $\gamma^{-1}(Y)$ is also open in $[0,1]$, then we will have shown that $[0,1]$ is disconnected, which is a contradiction.

To show that $\gamma^{-1}(Y)$ is open in $[0,1]$, it suffices to prove that around every point $t \in \gamma^{-1}(Y)$, there is $\varepsilon > 0$ such that $(t-\varepsilon, t+\varepsilon) \subseteq \gamma^{-1}(Y)$. This is equivalent to proving that $\gamma((t-\varepsilon, t+\varepsilon)) \subseteq Y$.

Let $t \in \gamma^{-1}(Y)$. Choose $\varepsilon > 0$ such that $\gamma((t-\varepsilon, t+\varepsilon))$ is contained in the closed disk D of radius $1/2$ with center $\gamma(t)$. The intersection $D \cap X$ contains a closed interval $D \cap Y$ on the y-axis, together with a number of segments of the curve $y = \sin(\pi/x)$, each of which is homeomorphic to a closed interval. Moreover, any two such segments are separated from one another in $D \cap X$. Therefore, $D \cap Y$ is a connected component of $D \cap X$. Since $\gamma(t) \in D \cap Y$, and $(t-\varepsilon, t+\varepsilon)$ is connected, we must have that $\gamma((t-\varepsilon, t+\varepsilon))$ is connected as well, and therefore $\gamma((t-\varepsilon, t+\varepsilon)) \subseteq D \cap Y$.

The previous paragraph argues that for any $t \in \gamma^{-1}(Y)$, there is $\varepsilon > 0$ such that $(t-\varepsilon, t+\varepsilon) \subseteq \gamma^{-1}(Y)$. Hence, $\gamma^{-1}(Y)$ is open. Since Y is closed in X, we also know that $\gamma^{-1}(Y)$ is closed as well. Since $0 \in \gamma^{-1}(Y)$ and $1 \notin \gamma^{-1}(Y)$, we have a nonempty proper open and closed subset of $[0,1]$, which contradicts the fact that $[0,1]$ is connected.

Therefore, X cannot be path connected; you cannot join a point in Y to a point in X by a path.

3.3.1 Path Components

Let X be a topological space. We do not assume that X is or is not path connected. Define an equivalence relation $\sim$ on points of X by $x \sim y$ if and only if there is a path $\gamma\colon [0,1] \to X$ that starts at x ($\gamma(0) = x$) and ends at y ($\gamma(1) = y$).

Definition 3.3.12. The *path components* of X are the equivalence classes of points in X the equivalence relation $\sim$ described above. We write $\pi_0(X)$ for the set of path components of X.

Example 3.3.13. If X is a path-connected space, then $\pi_0(X) = \{X\}$. This is a set with one element, and that element is the set X.

Example 3.3.14. Let $X = \mathbb{R} \setminus \{0\}$ with the subspace topology. Then there are two path components: the positive numbers and the negative numbers. Therefore, $\pi_0(X) = \{\mathbb{R}_{<0}, \mathbb{R}_{>0}\}$.

Example 3.3.15. Since between every two rational numbers you can find an irrational number, each rational number $a \in \mathbb{Q}$ is its own path component. Therefore, $\pi_0(\mathbb{Q}) = \{\{a\} \mid a \in \mathbb{Q}\}$.

Remark 3.3.16. A space is path-connected if and only if it has only one path component.

Chapter 4

Homotopy

Topological properties like compactness and connectedness have been pretty useful so far; they've let us show that $[0,1]$ and $(0,1)$ are not homeomorphic, because the latter is not compact but the former is. Similarly, we saw that $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic, because removing a point from $\mathbb{R}$ disconnects it, but removing a point from $\mathbb{R}^2$ does not. Compactness and connectedness are also essential to prove the extreme value theorem and the intermediate value theorem, which are super useful in their own right.

But we've barely scratched the surface. For example, we can't tell whether or not $\mathbb{R}^2$ and $\mathbb{R}^3$ are homeomorphic using any of the topological properties we've learned thus far. Similarly, we haven't justified that a torus T is not homeomorphic to a sphere S^2, nor that a torus is not homeomorphic to $\mathbb{RP}^2$ or a Klein bottle.

However, I'm willing to bet that you've seen a topological property in Calc 3 that *will* distinguish between $\mathbb{R}^3$ and $\mathbb{R}^2$, however briefly. When you studied line integrals, you probably talked about spaces that are *simply connected*. We'll make this rigorous later, but roughly speaking, a space is simply connected if every closed loop in the space can be shrunk to a point. Deleting a point in $\mathbb{R}^2$ leaves a space that is not simply connected, but deleting a point in $\mathbb{R}^3$ leaves a space that is simply connected. Therefore, they cannot be homeomorphic.

To introduce the idea of simply connected, we introduce a stronger invariant of spaces called the *fundamental group*. If two spaces are homeomorphic, then they have isomorphic fundamental groups. Therefore, we can prove that a sphere S^2 and a torus T are not homeomorphic by calculating their fundamental groups: it will turn out that the fundamental group of S^2 is trivial, but the fundamental group of the torus is not.

The fundamental group is important for much much more than just distinguishing spaces. We can use the ideas that we develop here to prove other important theorems: the fundamental theorem of algebra, and the Jordan curve theorem.

4.1 Homotopy of Paths

To properly define the fundamental group, we must first discuss how to continuously deform one path in a space into another path.

Definition 4.1.1. Let X be a space. Let $\alpha\colon [0,1] \to X$ and $\beta\colon [0,1] \to X$ be two paths in X.

We say that α and β are *path homotopic* if $\alpha(0) = \beta(0)$ (they start at the same point, call it x_0), $\alpha(1) = \beta(1)$

(they end at the same point, call it x_1), and there is a continuous map $H\colon [0,1]\times[0,1]\to X$ such that

$$\begin{aligned} H(s,0) &= \alpha(s),\\ H(s,1) &= \beta(s),\\ H(0,t) &= x_0,\\ H(1,t) &= x_1, \end{aligned}$$

for all $s,t\in[0,1]$. The map H is a *path homotopy* between α and β.

Example 4.1.2. Let $X=\mathbb{R}^2\setminus\{\vec{0}\}$, and consider the following two paths from $\vec{x}_0=(-1,0)$ to $\vec{x}_1=(1,0)$:

$$\begin{aligned} \alpha(s) &= (\cos(\pi s),\sin(\pi s))\\ \beta(s) &= (\cos(\pi s),2\sin(\pi s)). \end{aligned}$$

There is a path homotopy between these given by

$$H(s,t)=(1-t)\alpha(s)+t\beta(s).$$

This is an example of a *straight line homotopy*.

Example 4.1.3. In the same setup as the previous example with $X=\mathbb{R}^2\setminus\{\vec{0}\}$, consider the path

$$\gamma(s)=(\cos(\pi s),-\sin(\pi s)).$$

The straight line homotopy between α and γ is no longer allowed, because its image does not lie entirely inside the space $\mathbb{R}^2\setminus\{\vec{0}\}$. In fact, there is no path homotopy between α and γ in this setup. Hopefully it's intuitive that one cannot stretch and bend α around the "hole" at $\vec{0}$ while keeping the endpoints fixed at $(\pm1,0)$. We can't prove it yet, but we'll return to it later.

Another comment here: whether or not two paths are homotopic depends on the space X: α and γ would be homotopic if they were paths in $\mathbb{R}^2$.

Definition 4.1.4. Let X be a space. Let $\alpha,\beta\colon[0,1]\to X$ be two paths in X. If α and β are path homotopic, we write $\alpha\simeq\beta$. We write $[\alpha]$ for the path-homotopy class of α under this equivalence relation.

Proposition 4.1.5. *The relation $\simeq$ is an equivalence relation on paths in X.*

Proof. To check that this is an equivalence relation, we must check that it is reflexive, symmetric, and transitive.

To check reflexivity, let $\alpha\colon[0,1]\to X$ be a path. Then we can define a homotopy $H\colon[0,1]\times[0,1]\to X$ by

$$H(s,t)=\alpha(s).$$

This is a path homotopy from α to α, so $\alpha\simeq\alpha$.

To check symmetry, suppose that $\alpha\simeq\beta$. We want to show that $\beta\simeq\alpha$. Since $\alpha\simeq\beta$, there is a homotopy $H\colon[0,1]\times[0,1]\to X$ such that:

$$\begin{aligned} H(s,0) &= \alpha(s),\\ H(s,1) &= \beta(s), \end{aligned}$$

Then we can reverse the t-coordinate so to get a homotopy from β to α: let $\bar{H}(s,t)=H(s,1-t)$. Then

$$\begin{aligned} \bar{H}(s,0) &= H(s,1-0)=\beta(s),\\ \bar{H}(s,1) &= H(s,1-1)=\alpha(s), \end{aligned}$$

so $\bar{H}$ is the homotopy that we seek. Thus $\beta \simeq \alpha$.

Finally, to check transitivity, suppose that $\alpha \simeq \beta$ via a homotopy H_1 and that $\beta \simeq \gamma$ via a homotopy H_2. Then we seek a homotopy H from α to γ. That is, we seek a continuous function $H\colon [0,1] \times [0,1] \to X$ such that

$$\begin{aligned} H(s,0) &= \alpha(s), \\ H(s,1) &= \gamma(s), \\ H(0,t) &= x_0, \\ H(1,t) &= x_1. \end{aligned}$$

To construct this homotopy, let

$$H(s,t) = \begin{cases} H_1(s,2t) & 0 \le t \le \frac{1}{2} \\ H_2(s,2t-1) & = \frac{1}{2} \le t \le 1. \end{cases}$$

This map is well-defined because when $t = \frac{1}{2}$, we have

$$\begin{aligned} H_1(s,2(\tfrac{1}{2})) = H_1(s,1) = \beta(s) \\ H_2(s,2(\tfrac{1}{2})-1) = H_2(s,0) = \beta(s), \end{aligned}$$

so the pieces agree on the overlaps. Therefore, H is continuous by the pasting lemma (remember that one?), because both H_1 and H_2 are continuous and they agree on the overlap of the domains. By construction,

$$\begin{aligned} H(s,0) &= H_1(s,2(0)) = H_1(s,0) = \alpha(s) \\ H(s,1) &= H_2(s,2(1)-1) = H_2(s,1) = \gamma(s), \end{aligned}$$

and

$$H(0,t) = \begin{cases} H_1(0,2t) = x_0 & t \le \frac{1}{2}, \\ H_2(0,2t-1) = x_0 & t \ge \frac{1}{2}, \end{cases}$$

$$H(1,t) = \begin{cases} H_1(1,2t) = x_1 & t \le \frac{1}{2}, \\ \\ H_2(1,2t-1) = x_1 & t \ge \frac{1}{2}. \end{cases}$$

Therefore, H is the desired homotopy that witnesses $\alpha \simeq \gamma$. □

4.1.1 Products of Paths

We are now ready to introduce our first bit of algebra into the mix. This will be the group operation in the fundamental group, when we get there.

Definition 4.1.6. If $\alpha\colon [0,1] \to X$ is a path from x_0 to x_1, and $\beta\colon [0,1] \to X$ is a path from x_1 to x_2, then we define the *product* $\alpha * \beta$ of α and β to be the path defined by the equation

$$(\alpha * \beta)(s) = \begin{cases} \alpha(2s) & 0 \le s \le \frac{1}{2}, \\ \beta(2s-1) & \frac{1}{2} \le s \le 1. \end{cases}$$

Lemma 4.1.7. *The product of paths is a well-defined operation on equivalence classes of paths: if $\alpha \simeq \alpha'$ and $\beta \simeq \beta'$, then $\alpha * \beta \simeq \alpha' * \beta'$.*

Proof. Let H_1 be a path homotopy between α and α', and let H_2 be a path homotopy between β and β'. Then we may define

$$H(s,t) = \begin{cases} H_1(2s,t) & 0 \le s \le \frac{1}{2}, \\ H_2(2s-1,t) & \frac{1}{2} \le s \le 1. \end{cases}$$

You can check that H is well-defined, continuous, and is the required path homotopy between $\alpha * \beta$ and $\alpha' * \beta'$. □

Knowing this, we have a well-defined product operation on path-homotopy classes defined by

$$[\alpha] * [\beta] = [\alpha * \beta].$$

This operation will work very much like a group operation, even though it can't be a group operation because it's only defined for homotopy classes of paths $[\alpha] * [\beta]$ when the endpoint of α is the start point of β. A group operation must be defined for all elements of the group. This is something weirder called a *groupoid*.

Theorem 4.1.8. *The operation $*$ on homotopy classes of paths in X has the following properties:*

(a) (Right/Left Identity) Given $x \in X$, let $e_x : [0,1] \to X$ be the constant path defined by $e_x(t) = x$ for all $t \in [0,1]$. Then if $\alpha : [0,1] \to X$ is a path from x_0 to x_1, then

$$[\alpha] * [e_{x_1}] = [\alpha] \text{ and } [e_{x_0}] * [\alpha] = [\alpha].$$

(b) (Inverse) Given a path $\alpha : [0,1] \to X$ from x_0 to x_1, let $\bar{\alpha}$ be the path defined by $\bar{\alpha}(t) = \alpha(1-t)$. It is called the reverse of α. Then

$$[\bar{\alpha}] * [\alpha] = [e_{x_1}] \text{ and } [\alpha] * [\bar{\alpha}] = [e_{x_0}].$$

(c) (Associativity) If $[\alpha] * ([\beta] * [\gamma])$ is defined, then so is $([\alpha] * [\beta]) * [\gamma]$ and they are equal.

The proof of this theorem will occupy the next several classes. We first need to prove a few lemmas.

Lemma 4.1.9. *If $f : X \to Y$ is a continuous map, and H is a path homotopy between paths $\alpha : [0,1] \to X$ and $\beta : [0,1] \to X$, then $f \circ H$ is a path homotopy between paths $f \circ \alpha$ and $f \circ \beta$ in Y.*

Proof. We must check the conditions of a path homotopy in Y. □

Lemma 4.1.10. *If $f : X \to Y$ is a continuous map and if $\alpha : [0,1] \to X$ is a path from x_0 to x_1, and $\beta : [0,1] \to X$ is a path from x_1 to x_2, then we have*

$$f \circ (\alpha * \beta) = (f \circ \alpha) * (f \circ \beta).$$

Proof. This follows directly from the definition of the operation $*$. □

Next, we check the identity property of the product of paths.

Proposition 4.1.11. *Given $x \in X$, let $e_x : [0,1] \to X$ be the constant path defined by $e_x(t) = x$ for all $t \in [0,1]$. Then if $\alpha : [0,1] \to X$ is a path from x_0 to x_1, then*

$$[\alpha] * [e_{x_1}] = [\alpha] \text{ and } [e_{x_0}] * [\alpha] = [\alpha].$$

Proof. Consider the identity map $\mathrm{id} : [0,1] \to [0,1]$, $\mathrm{id}(t) = t$. This is a path in $[0,1]$ if we interpret it that way. There is another path in $[0,1]$ given by the constant path at zero, $e_0 : [0,1] \to [0,1]$, $e_0(t) = 0$. Then we can take the product of these two paths, $e_0 * \mathrm{id}$, to get a new path from $[0,1]$ to itself, given by

$$(e_0 * \mathrm{id})(t) = \begin{cases} 0 & t \le \frac{1}{2} \\ 2t-1 & t \ge \frac{1}{2}. \end{cases}$$

Since the unit interval $[0,1]$ is convex, there is a path homotopy H in $[0,1]$ from id and $e_0 * \mathrm{id}$.

Then if $\alpha\colon [0,1] \to X$ is a path in X from x_0 to x_1, we can use Lemmas 4.1.9 and 4.1.10 and compose α with the homotopy H to get a new homotopy between $\alpha \circ \mathrm{id} = \alpha$ and

$$\alpha \circ (e_0 * \mathrm{id}) = (\alpha \circ e_0) * (\alpha \circ \mathrm{id}) = e_{x_0} * \alpha.$$

Therefore, $\alpha \simeq e_{x_0} * \alpha$, so α has a left identity. Similarly, we can show that $\alpha \simeq \alpha * e_{x_1}$. □

Proposition 4.1.12. *Given a path $\alpha\colon [0,1] \to X$ from x_0 to x_1, let $\bar{\alpha}$ be the path defined by $\bar{\alpha}(t) = \alpha(1-t)$. It is called the reverse of α. Then*

$$[\bar{\alpha}] * [\alpha] = [e_{x_1}] \text{ and } [\alpha] * [\bar{\alpha}] = [e_{x_0}].$$

Proof. Consider again the identity map $\mathrm{id}\colon [0,1] \to [0,1]$ given by $\mathrm{id}(t) = t$. The reverse of this map is $\bar{\mathrm{id}}$, given by $\bar{\mathrm{id}}(t) = \mathrm{id}(1-t) = 1-t$. Then $\mathrm{id} * \bar{\mathrm{id}}$ is a path in $[0,1]$ that both begins and ends at 0. Another path that both begins and ends at 0 is the constant path e_0. Because $[0,1]$ is convex, there is a path homotopy H in $[0,1]$ between e_0 and $\mathrm{id} * \bar{\mathrm{id}}$.

Now let $\alpha\colon [0,1] \to X$ be a path in X from x_0 to x_1. Using Lemmas 4.1.9 and 4.1.10, we find that $\alpha \circ H$ is a path homotopy between $\alpha \circ e_0 = e_{x_0}$ and

$$(\alpha \circ \mathrm{id}) * (\alpha \circ \bar{\mathrm{id}}) = \alpha * \bar{\alpha}.$$

Therefore, $\alpha * \bar{\alpha} \simeq e_{x_0}$. A similar argument, using the fact that $\bar{\mathrm{id}} * \mathrm{id}$ is path homotopic to e_1, shows that $\bar{\alpha} * \alpha = e_{x_1}$. □

Proposition 4.1.13. *If $[\alpha] * ([\beta] * [\gamma])$ is defined, then so is $([\alpha] * [\beta]) * [\gamma]$ and they are equal.*

Proof. This is the hardest of the three. Consider the map $f\colon [0,1] \to [0,1]$ given by

$$f(s) = \begin{cases} 2s & 0 \le s \le \frac{1}{4}, \\ s + \frac{1}{4} & \frac{1}{4} \le s \le \frac{1}{2}, \\ \frac{s+1}{2} & \frac{1}{2} \le s \le 1. \end{cases}$$

We claim that $f\colon [0,1] \to [0,1]$ is path-homotopic to $\mathrm{id}\colon [0,1] \to [0,1]$. Indeed,

$$H(s,t) = (1-t)\mathrm{id}(s) + tf(s)f$$

is the desired homotopy. Then we have

$$(\alpha * \beta) * \gamma = (\alpha * (\beta * \gamma)) \circ f \simeq (\alpha * (\beta * \gamma)) \circ \mathrm{id} = \alpha * (\beta * \gamma).$$

□

Proof of Theorem 4.1.8. This is proved in Propositions 4.1.11 to 4.1.13. □

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